

RECIPROCAL SUMS OVER SEPARATED INTEGER INTERVALS

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ABSTRACT. We prove that for every sufficiently large integer k there is a finite set $S \subset \mathbb{N}_{>0}$ that is the disjoint union of exactly k pairwise nonadjacent integer intervals, each of cardinality 2 or 3, and satisfies $\sum_{n \in S} 1/n = 1$. This gives an affirmative answer, in a slightly stronger bounded-length form, to the strengthened disjoint, nonadjacent formulation currently listed as Erdős Problem 289. The proof follows the prime-power filtration of $\mathbb{Q}/\mathbb{Z}$: signed-inverse interval families provide controlled local residues, while reciprocal-neutral identities provide freedom in the number of components. Restricted-sum estimates yield eventual coverage of all sufficiently large component counts; after a finite cutoff, Haxell’s independent-transversal theorem realizes the local choices by mutually compatible intervals. Affine-fibre pullbacks and a compatible covering theorem assemble these inputs. A final mass bound lifts residue zero to the exact reciprocal value 1; all three-element components may be confined to a fixed finite region.

1. INTRODUCTION

We use $\mathbb{N} = \mathbb{Z}_{\geq 0}$ and $\mathbb{N}_{>0} = \{1, 2, \dots\}$. Let $\mathcal{C}$ be the family of finite subsets $S \subset \mathbb{N}_{>0}$ whose connected components in the path graph have cardinality 2 or 3. Write

$$\nu(S) = |\pi_0(S)|, \quad W(S) = \sum_{n \in S} \frac{1}{n}.$$

Theorem 1.1. *There is $k_0 \in \mathbb{N}_{>0}$ such that for every $k \geq k_0$ there exists $S \in \mathcal{C}$ with*

$$\nu(S) = k, \quad W(S) = 1.$$

Equivalently, every sufficiently large k is realized by exactly k pairwise disjoint, nonadjacent integer intervals, each of cardinality 2 or 3, whose total reciprocal sum is 1.

This gives an affirmative answer to the strengthened formulation listed as Erdős Problem 289 [4], with the additional restriction that each interval has cardinality 2 or 3. Erdős and Graham posed the underlying interval-reciprocal problem in [7, p. 34]; their printed formulation does not explicitly impose distinctness, disjointness, or nonadjacency. We use the strengthened formulation recorded in the current problem list. A closely related Monthly problem of Hahn [8, p. 47] asked for integral reciprocal sums of disjoint intervals. In the published solution

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to E2689, Montgomery gave the separated five-interval representation of 2 [9, p. 224]. Erdős and Graham reproduce the representation and record Hickerson's independent solution [7, p. 34].

The surrounding unit-fraction literature studies different forms of localization and largeness. Martin constructed dense Egyptian-fraction representations [11], Croot studied representations with denominators in short intervals [5], and Bloom proved the positive-upper-density conjecture of Erdős and Graham [3]. These results do not impose the separated-block component constraint considered here.

The construction gives a further restriction on the three-element intervals.

Corollary 1.2. *There are integers $M, C, k_0 \geq 1$ such that, for every $k \geq k_0$, the set S in Theorem 1.1 can be chosen with every three-element component contained in $[1, M]$ and*

$$0 \leq |S| - 2k \leq C.$$

Thus three-element intervals are needed only in a fixed finite part of the construction. The corollary is proved at the end of Section 9.

The structural argument. The proof separates observations from their witnesses. The observation of a configuration is

$$\Omega(S) = (\overline{W}(S), \nu(S)) \in (\mathbb{Q}/\mathbb{Z}) \times \mathbb{N}, \quad \overline{W}(S) = [W(S)].$$

It is additive under compatible union. Two difficulties remain invisible at this level: local residue choices may require different component counts, and separately valid choices may use overlapping or adjacent intervals.

We retain the interval configurations as labels on finite correspondences. A covering correspondence has a surjective target map from its witness object. Local residue-count responses lift to covering correspondences by an affine-fibre pullback. Their finite composite is another pullback, restricted to the domain of joint compatibility. The compatible covering theorem, Theorem 2.4, says that this composite is covering once its path-label map factors through that domain. These universal properties organize the proof; the arithmetic below verifies their hypotheses.

Here is the final base change, with actual sets and maps. At a finite cutoff X , write $\mathcal{E}_X$ for the witness object of a covering correspondence with target $T_X \subseteq (\mathbb{Q}/\mathbb{Z}) \times \mathbb{N}$, and let $I_X^+ = I_X \cap \mathbb{N}_{>0}$. Once the affine centre has vanished, $(0, k) \in T_X$ for every $k \in I_X^+$. The left square in

$$\begin{array}{ccccc} \mathcal{E}_X^0 & \longrightarrow & \mathcal{E}_X & \xrightarrow{\ell_X} & C \\ \pi_X \downarrow & & \downarrow t_X & & \downarrow \Omega \\ I_X^+ & \xrightarrow{k \mapsto (0, k)} & T_X & \longrightarrow & (\mathbb{Q}/\mathbb{Z}) \times \mathbb{N} \end{array} \quad (1.1)$$

is a pullback, and the right square is the conservation law. Pullback stability makes π_X surjective. A mass bound $W(\ell_X(e)) < 2$ then makes every label over the lower left corner an exact solution: its reciprocal value is a positive integer below 2.

For

$$\mathcal{S} = \{(k, S) : k \in \mathbb{N}_{>0}, S \in C, \nu(S) = k, W(S) = 1\}, \quad \pi_{\text{sol}}(k, S) = k, \quad (1.2)$$

this gives $I_X^+ \subseteq \text{im}(\pi_{\text{sol}})$. The remaining task is to construct these covers with a uniform mass bound and to show that the intervals I_X^+ eventually cover a final ray.

Arithmetic and compatibility inputs. The residue filtration is

$$E_X = H_{\text{lcm}(1, \dots, \lfloor X \rfloor)} \subseteq \mathbb{Q}/\mathbb{Z}, \quad H_n = \{x : nx = 0\}.$$

Its nontrivial steps occur at prime powers $Q = p^e$, and each consecutive quotient has order p . This reduces residue transfer to a succession of simple cyclic quotients, without choosing a simultaneous system of coordinates for the whole torsion group.

A finite *prefix* covers an affine E_B -fibre with bounded spread in component count. Reciprocal-neutral two-versus-three component switches widen that spread to a prescribed interval of width $o(B/\log B)$, at arbitrarily small additional mass. A sparse-interaction estimate ensures that this prefix leaves a fixed positive proportion of every later interval family available, uniformly in the placement of the neutral switches.

At each later prime-power step, signed inverses produce $\gg Q/\log Q$ binary intervals with controlled quotient values, bounded incompatibility degree and summable role mass. The Dias da Silva–Hamidoune restricted-sum theorem gives a residue-count rectangle when the quotient image is large; otherwise a heavy fibre gives the required response. These statements hold uniformly after every sufficiently dense restriction of a row. Prime steps create width $\gg q/\log q$, whereas the cumulative width loss from composite prime powers up to Q is $o(Q/\log Q)$. A first-failure argument therefore gives overlapping count intervals with unbounded upper endpoints.

Only after a finite cutoff is fixed do we apply Haxell’s independent-transversal theorem. It provides one independent family retaining a fixed positive proportion of every active row. Uniformity under dense restriction preserves the local responses, while independence supplies the compatibility factorization required by Theorem 2.4. Finally, the cofinality $\bigcup_X E_X = \mathbb{Q}/\mathbb{Z}$ kills the fixed affine centre, and (1.1) completes the proof.

Section 2 develops the categorical transfer calculus. Section 3 records the four auxiliary mechanisms used to construct its inputs. Sections 4–6 establish the arithmetic families, the prefix and the local responses. Sections 7–9 prove count coverage, impose finite compatibility and take the final base change. The appendices give the prime-supply estimate, a polynomial proof of the restricted-sum theorem and the exact finite initialization.

2. PHYSICAL CORRESPONDENCES, AFFINE FIBRES, AND TORSION

The structural part of the proof has three ingredients: compatible interval configurations with additive observations, finite labelled correspondences composed by pullback over the literal compatibility domain, and affine fibres in the prime-power filtration of $\mathbb{Q}/\mathbb{Z}$. The constructions below are used only where a universal property controls a later transport or pasting step.

Let C be the set of finite subsets $S \subset \mathbb{N}_{>0}$ whose connected components in the path graph have cardinality 2 or 3. Put

$$W(S) = \sum_{n \in S} \frac{1}{n}, \quad \nu(S) = |\pi_0(S)|, \quad \overline{W}(S) = [W(S)] \in A := \mathbb{Q}/\mathbb{Z}. \quad (2.1)$$

Let

$$M = A \times \mathbb{N}, \quad 0_M = (0, 0), \quad \Omega = (\overline{W}, \nu) : C \longrightarrow M.$$

For $n \geq 1$, let

$$D_n \subseteq C^n$$

be the joint-compatibility domain: $(S_1, \dots, S_n) \in D_n$ when the supports are pairwise disjoint and their literal union is again an element of C . Write $\mu_n : D_n \rightarrow C$ for literal union. Hence

$$W\left(\bigcup_i S_i\right) = \sum_i W(S_i), \quad \nu\left(\bigcup_i S_i\right) = \sum_i \nu(S_i), \quad \overline{W}\left(\bigcup_i S_i\right) = \sum_i \overline{W}(S_i). \quad (2.2)$$

Equivalently, for binary compatible unions the square

$$\begin{array}{ccc} D_2 & \xrightarrow{\mu_2} & C \\ (\Omega\pi_1, \Omega\pi_2) \downarrow & & \downarrow \Omega \\ M^2 & \xrightarrow{+} & M \end{array} \quad (2.3)$$

commutes.

Lemma 2.1 (Compatibility substitution law). *Every subtuple of a point of D_n is jointly compatible. More generally, if the indices are partitioned into blocks, taking the union within each block gives a compatible tuple of block unions, and its final union is μ_n . Conversely, every iterated binary-compatible parenthesization determines a point of D_n . Thus all parenthesizations have the common domain D_n and the common value μ_n .*

Proof. Suppose $(S_1, \dots, S_n) \in D_n$. If a component of S_i were adjacent to a component of S_j with $i \neq j$, those two components would lie in one component of the total union and that component would have cardinality at least 4, contrary to the definition of C . Components belonging to different labels are therefore separated by at least one unused integer. Every subunion and every block union is consequently again an element of C , and substitution does not change the literal total union.

Conversely, an iterated binary-compatible product records pairwise disjoint labels whose final literal union belongs to C ; these are exactly the defining conditions for membership in D_n . $\square$

Thus C , with unit $\emptyset$, domains D_n and partial products μ_n , is a partial commutative monoid, and Ω is additive wherever the partial product is defined.

Remark 2.2 (Abstract scope). Everything in the next two subsections is formal for a partial commutative monoid $\mathcal{C}$ equipped with an additive map $\Omega : \mathcal{C} \rightarrow M$ to a commutative monoid and a nonnegative additive mass W . The interval model enters only through the verification of the partial-sum axioms and, later, through the arithmetic estimates and the incompatibility graph.

2.1. Witnesses and regular images. All pullbacks and images below are taken in $\mathbf{Set}$. A regular epimorphism is therefore a surjection; its stability under pullback and composition is the mechanism that transports existence. We retain a finite witness object E and its target map $E \twoheadrightarrow Y$ until compatibility has been imposed. Passing to an image records which targets occur but may discard the information needed to compose their witnesses.

2.2. Labelled correspondences and compatible pasting. Let $X, Y \subseteq M$ be finite. A labelled finite correspondence

$$R : X \rightsquigarrow Y$$

consists of a finite edge object E_R and maps

$$s_R : E_R \rightarrow X, \quad \ell_R : E_R \rightarrow C, \quad t_R : E_R \rightarrow Y$$

satisfying the observable conservation law

$$t_R(e) = s_R(e) + \Omega(\ell_R(e)) \quad (e \in E_R). \quad (2.4)$$

Equivalently, the square

$$\begin{array}{ccc} E_R & \xrightarrow{(s_R, \ell_R)} & X \times C \\ t_R \downarrow & & \downarrow (x, L) \mapsto x + \Omega(L) \\ Y & \longrightarrow & M \end{array} \quad (2.5)$$

commutes, where the bottom arrow is inclusion. The correspondence is *covering* when $t_R : E_R \twoheadrightarrow Y$ is a regular epimorphism.

For correspondences $R : X \rightsquigarrow Y$ and $S : Y \rightsquigarrow Z$, first form the path pullback

$$\begin{array}{ccc} P(S, R) & \xrightarrow{\text{pr}_S} & E_S \\ \text{pr}_R \downarrow & & \downarrow s_S \\ E_R & \xrightarrow{t_R} & Y. \end{array} \quad (2.6)$$

The pair of label maps gives

$$(\ell_R \text{pr}_R, \ell_S \text{pr}_S) : P(S, R) \rightarrow C^2.$$

The compatible path object is the pullback

$$\begin{array}{ccc} P^{\text{comp}}(S, R) & \longrightarrow & D_2 \\ \downarrow & & \downarrow \\ P(S, R) & \xrightarrow{(\ell_R \text{pr}_R, \ell_S \text{pr}_S)} & C^2. \end{array} \quad (2.7)$$

Define $S \odot R$ to have edge object $P^{\text{comp}}(S, R)$, source $s_R \text{pr}_R$, target $t_S \text{pr}_S$, and label obtained from the upper map in (2.7) followed by union $D_2 \rightarrow C$. The conservation square for $S \odot R$ is the pasted composite of (2.3), (2.6), and the two conservation squares of R and S .

The identity correspondence on X has edge object X and empty label. A 2-cell is a map of edge objects preserving source, target and label; vertical composition is composition of these edge maps. If $\alpha : R \Rightarrow R'$ and $\beta : S \Rightarrow S'$, the induced map $P(S, R) \rightarrow P(S', R')$ restricts, by the universal property of the compatibility pullback, to $P^{\text{comp}}(S, R) \rightarrow P^{\text{comp}}(S', R')$; this is horizontal composition. This

is the only higher structure required here: it records changes of witnesses without changing their observations or their physical labels.

Lemma 2.3. *Compatible pasting makes the finite state sets and labelled correspondences into a bicategory $\mathbf{Corr}_C^\Omega$. More precisely, both parenthesizations of any finite composite are canonically represented by the same compatible path object*

$$P_n^{\text{comp}} = P_n \times_{C^n} D_n,$$

where P_n is the iterated fibre product of the edge objects over the intermediate state sets. We henceforth suppress these canonical pullback associators.

Proof. The iterated path object P_n is characterized by the universal property of the fibre product, independently of parenthesization. Pulling it back along $D_n \hookrightarrow C^n$ imposes joint compatibility in one step. Lemma 2.1 shows that every intermediate block union required by any parenthesization is defined on this same object and has total label μ_n . The identity statement follows from the empty label. The canonical pullback isomorphisms satisfy the usual coherence because they are uniquely determined by their projections. $\square$

The underlying unlabelled relation is the regular image

$$U(R) = \text{im}(s_R, t_R) \subseteq X \times Y. \quad (2.8)$$

A 2-cell $R \Rightarrow R'$ induces the inclusion $U(R) \subseteq U(R')$, which is the action of U on 2-cells. For $E_R \neq \emptyset$ put

$$\omega(R) = \max\{W(\ell_R(e)) : e \in E_R\},$$

and set $\omega(R) = 0$ when $E_R = \emptyset$.

Since (2.7) is a subobject of the unrestricted path pullback, every composable pair has a canonical comparison

$$U(S \odot R) \subseteq U(S) \circ U(R), \quad (2.9)$$

The bicategory $\mathbf{Rel}$ has sets as objects, relations as arrows and inclusions as 2-cells. Thus U is a normal oplax homomorphism to $\mathbf{Rel}$, with the comparison direction displayed in (2.9). The possible failure of equality has a concrete meaning: two successive edges may reuse the same nonempty interval label. Their endpoint observations compose, but their labels do not.

For a finite composable string $R_1, \dots, R_n$, let

$$\pi_{0,n} : P_n \longrightarrow X_0 \times X_n$$

be the endpoint map and let $\ell : P_n \rightarrow C^n$ be the label map. Then

$$U(R_n \odot \dots \odot R_1) = \text{im}(\pi_{0,n}|_{P_n^{\text{comp}}}), \quad U(R_n) \circ \dots \circ U(R_1) = \text{im}(\pi_{0,n}). \quad (2.10)$$

Hence the comparison is an identity precisely when

$$\text{im}(\pi_{0,n}|_{P_n^{\text{comp}}}) = \text{im}(\pi_{0,n}). \quad (2.11)$$

The information loss is displayed by the commuting image square

$$\begin{array}{ccc} P_n^{\text{comp}} & \xrightarrow{\quad} & P_n \\ \pi_{0,n} \downarrow & & \downarrow \pi_{0,n} \\ U(R_n \odot \dots \odot R_1) & \longrightarrow & U(R_n) \circ \dots \circ U(R_1), \end{array} \quad (2.12)$$

where the lower objects denote the corresponding regular images. A stronger witnesswise condition is that the path-label map factor through the compatibility domain:

$$P_n \xrightarrow{\tilde{\ell}} D_n \xrightarrow{\iota} C^n, \quad \ell = \iota \circ \tilde{\ell}. \quad (2.13)$$

By the universal property of (2.7), this factorization is equivalent to the canonical map

$$P_n^{\text{comp}} \longrightarrow P_n$$

being an isomorphism. It therefore implies

$$U(R_n \odot \cdots \odot R_1) = U(R_n) \circ \cdots \circ U(R_1). \quad (2.14)$$

Moreover, for every composable string,

$$\omega(R_n \odot \cdots \odot R_1) \leq \sum_{i=1}^n \omega(R_i), \quad (2.15)$$

because the label of a compatible path is the literal union of its labels and reciprocal value is additive there.

Theorem 2.4 (Compatible covering theorem). *Let $R_i : X_{i-1} \rightsquigarrow X_i$ be a finite string of covering labelled correspondences. If its path-label map factors through D_n , then the compatible composite is covering. Its target map is obtained inductively from the local target maps by pullback and composition, and its label mass is at most $\sum_i \omega(R_i)$.*

Proof. Let P_i be the path object through R_i , and let $\tau_i : P_i \rightarrow X_i$ be its target map. For $i \geq 2$ the square

$$\begin{array}{ccc} P_i & \xrightarrow{\text{pr}_i} & E_{R_i} \\ \downarrow & & \downarrow s_i \\ P_{i-1} & \xrightarrow{\tau_{i-1}} & X_{i-1} \end{array}$$

is a pullback. Starting from $\tau_1 = t_1$, pullback stability makes pr_i a regular epimorphism, and composition makes $\tau_i = t_i \text{pr}_i$ a regular epimorphism. The compatibility factorization identifies P_n^{comp} with P_n . Under this identification its target is τ_n , and its label is $\mu_n \tilde{\ell}$. The mass assertion is (2.15). $\square$

The theorem separates two proof obligations. Quotient calculations must provide covering local correspondences; the physical construction must make their path labels factor through D_n . Once these obligations are met, the global existence statement follows by pasting pullbacks. Neither obligation can be replaced by the other.

2.3. Finite torsion and affine fibres in $\mathbb{Q}/\mathbb{Z}$. For $n \geq 1$ put

$$H_n = \{x \in A : nx = 0\}.$$

For a subgroup $H \leq A$, let

$$q_H : A \longrightarrow A/H$$

be the quotient map, and for $\alpha \in A/H$ write

$$\text{Fib}_H(\alpha) = q_H^{-1}(\alpha)$$

for the corresponding affine torsor fibre. When H is finite, this fibre is a finite coset of H . In particular, every set $\text{Fib}_H(\alpha) \times I$ used below is a finite state object.

2.4. Affine extension as a pullback of regular epimorphisms. Let $H \leq K \leq A$ be finite subgroups and write

$$\pi_{H,K} : A/H \longrightarrow A/K$$

for the quotient map. The subgroup inclusion is encoded by the exact sequence

$$0 \longrightarrow K/H \longrightarrow A/H \xrightarrow{\pi_{H,K}} A/K \longrightarrow 0 \quad (2.16)$$

and by the pullback square

$$\begin{array}{ccc} K & \longrightarrow & A \\ q_H|_K \downarrow & & \downarrow q_H \\ K/H & \longrightarrow & A/H. \end{array} \quad (2.17)$$

For $\beta \in A/K$ put

$$\text{Fib}_{\pi_{H,K}}(\beta) = \pi_{H,K}^{-1}(\beta) \subseteq A/H.$$

Translation by $\alpha \in A/H$ gives a canonical torsor isomorphism

$$\text{Fib}_{\pi_{H,K}}(\beta) \xrightarrow{u \mapsto \alpha + u} \text{Fib}_{\pi_{H,K}}(\pi_{H,K}(\alpha) + \beta). \quad (2.18)$$

Lemma 2.5 (Affine torsor extension principle). *Let $\alpha \in A/H$ and $\beta \in A/K$. Let $d : Y \rightarrow A$ be a map such that $q_K d$ is constantly β and*

$$q_H d : Y \twoheadrightarrow \text{Fib}_{\pi_{H,K}}(\beta) \quad (2.19)$$

is a regular epimorphism. Put

$$F^- = \text{Fib}_H(\alpha), \quad F^+ = \text{Fib}_K(\pi_{H,K}(\alpha) + \beta).$$

Then the square

$$\begin{array}{ccc} F^- \times Y & \xrightarrow{(x,y) \mapsto x + d(y)} & F^+ \\ \text{pr}_Y \downarrow & & \downarrow q_H \\ Y & \xrightarrow{y \mapsto \alpha + q_H d(y)} & \text{Fib}_{\pi_{H,K}}(\pi_{H,K}(\alpha) + \beta) \end{array} \quad (2.20)$$

is a pullback. Consequently its upper horizontal map is a regular epimorphism. More generally, if $r : X \twoheadrightarrow F^-$ is any regular epimorphism, then

$$X \times Y \longrightarrow F^+, \quad (x, y) \longmapsto r(x) + d(y)$$

is a regular epimorphism.

Proof. The lower arrow in (2.20) is the composite of the regular epimorphism (2.19) with the isomorphism (2.18). The square is a pullback: for $y \in Y$ and $z \in F^+$ with

$$q_H(z) = \alpha + q_H d(y),$$

the element $x = z - d(y)$ belongs to F^- and is the unique point mapping to (y, z) . Hence the upper map is the pullback of a regular epimorphism and is therefore a regular epimorphism. The final assertion follows by composing it with the product regular epimorphism $r \times 1_Y$. $\square$

Corollary 2.6 (Compatible affine extension). *Let $\mathcal{A}$ and $\mathcal{B}$ be finite parameter sets with label maps*

$$\lambda_{\mathcal{A}} : \mathcal{A} \rightarrow C, \quad \lambda_{\mathcal{B}} : \mathcal{B} \rightarrow C,$$

and suppose every pair of labels $\lambda_{\mathcal{A}}(s)$ and $\lambda_{\mathcal{B}}(b)$ is compatible. Suppose

$$\overline{W}\lambda_{\mathcal{A}} : \mathcal{A} \twoheadrightarrow \text{Fib}_H(\alpha)$$

is a regular epimorphism, $q_K \overline{W}\lambda_{\mathcal{B}}$ is constant with value $\beta \in A/K$, and

$$q_H \overline{W}\lambda_{\mathcal{B}} : \mathcal{B} \twoheadrightarrow \text{Fib}_{\pi_{H,K}}(\beta) \quad (2.21)$$

is a regular epimorphism. Then the compatible-union map

$$\mathcal{A} \times \mathcal{B} \longrightarrow C, \quad (s, b) \longmapsto \lambda_{\mathcal{A}}(s) \cup \lambda_{\mathcal{B}}(b)$$

has residue image exactly

$$\text{Fib}_K(\pi_{H,K}(\alpha) + \beta).$$

If $v\lambda_{\mathcal{B}}$ is constant with value c , the component-count set is translated by c , so its spread is unchanged.

Proof. Apply Lemma 2.5 to the regular epimorphism $\overline{W}\lambda_{\mathcal{A}}$ and the increment map $d = \overline{W}\lambda_{\mathcal{B}}$. Compatibility makes the union map defined on the whole product, and (2.3) identifies its residue map with addition. $\square$

Keeping the component count. Let $I, I' \subseteq \mathbb{N}$ be finite intervals, let $\alpha \in A/H$ and $\beta \in A/K$, and let $\mathcal{L}$ be a finite label-parameter set with label map $\lambda : \mathcal{L} \rightarrow C$. Suppose $q_K \overline{W}\lambda$ is constantly β . Form

$$Y_{I'} = \{(L, i) \in \mathcal{L} \times I : i + v(\lambda(L)) \in I'\}$$

and the response map

$$\begin{aligned} \Psi : Y_{I'} &\longrightarrow \text{Fib}_{\pi_{H,K}}(\pi_{H,K}(\alpha) + \beta) \times I', \\ (L, i) &\longmapsto (\alpha + q_H \overline{W}(\lambda(L)), i + v(\lambda(L))). \end{aligned} \quad (2.22)$$

Assume Ψ is a regular epimorphism and put

$$T^- = \text{Fib}_H(\alpha) \times I, \quad T^+ = \text{Fib}_K(\pi_{H,K}(\alpha) + \beta) \times I'.$$

Let

$$\chi : T^+ \longrightarrow \text{Fib}_{\pi_{H,K}}(\pi_{H,K}(\alpha) + \beta) \times I', \quad (z, k') \longmapsto (q_H(z), k').$$

Define the edge object by the pullback

$$\begin{array}{ccc} E_{\text{Tr}_{\mathcal{L}}} & \longrightarrow & Y_{I'} \\ \downarrow t & & \downarrow \Psi \\ T^+ & \xrightarrow{\chi} & \text{Fib}_{\pi_{H,K}}(\pi_{H,K}(\alpha) + \beta) \times I'. \end{array} \quad (2.23)$$

For a point $((z, k'), (L, i))$ of this pullback put

$$s((z, k'), (L, i)) = (z - \overline{W}(\lambda(L)), i), \quad \ell((z, k'), (L, i)) = \lambda(L).$$

Corollary 2.7 (Graded affine translation). *For the preceding pullback, s lands in T^- , and (s, ℓ, t) is a covering labelled correspondence*

$$\text{Tr}_{\mathcal{L}}(T^-, T^+) : T^- \rightsquigarrow T^+.$$

Proof. The pullback identity gives

$$q_H(z) = \alpha + q_H \overline{W}(\lambda(L)), \quad k' = i + v(\lambda(L)).$$

Hence $z - \overline{W}(\lambda(L)) \in \text{Fib}_H(\alpha)$, so the source is well defined, and the conservation identity is immediate. Since Ψ is a regular epimorphism, pullback stability makes t a regular epimorphism. $\square$

2.5. The prime-power filtration.

Lemma 2.8. *For $m, n \geq 1$,*

$$H_n = \langle [1/n] \rangle, \quad |H_n| = n,$$

$$H_m \leq H_n \iff m \mid n, \quad H_m \vee H_n = H_{\text{lcm}(m,n)}.$$

Consequently the nonzero join-irreducible stages are exactly H_{p^e} , with p prime and $e \geq 1$.

Proof. If $[q] \in H_n$, then $nq \in \mathbb{Z}$, so $[q] = [j/n]$ for some $j \in \mathbb{Z}$; hence $H_n = \langle [1/n] \rangle$ and has order n . The inclusion criterion follows by testing $[1/m]$. Put $L = \text{lcm}(m, n)$. Since $\gcd(L/m, L/n) = 1$, there exist $a, b \in \mathbb{Z}$ with

$$aL/m + bL/n = 1.$$

Dividing by L gives $[1/L] \in H_m + H_n$, whence $H_m \vee H_n = H_L$. The join-irreducibles of the divisibility lattice are precisely the prime powers. $\square$

Accordingly we index the nontrivial steps by prime powers $Q = p^e$. The *rank* of the stage H_{p^e} means the positive integer $Q = p^e$ itself. Let F_Q be the join of all prime-power stages of rank at most Q , and $F_{<Q}$ the join of those of smaller rank. The simple quotient

$$S_Q = F_Q / F_{<Q} \tag{2.24}$$

has order p : the p -adic valuation of the lower least common multiple is $e - 1$, and adjoining p^e raises it by one. Thus S_Q is a one-dimensional $\mathbb{F}_p$ -object.

For $X \geq 1$ define the endpoint stage

$$E_X = H_{\text{lcm}(1, \dots, [X])}. \tag{2.25}$$

Every class $[a/b] \in A$ belongs to H_b , so

$$A = \bigcup_{X \geq 1} E_X. \tag{2.26}$$

Equivalently, for every B one has the identity of subsets of A/E_B

$$A/E_B = \bigcup_{X \geq B} \ker(\pi_{B,X} : A/E_B \rightarrow A/E_X). \tag{2.27}$$

Indeed $\ker(\pi_{B,X}) = q_{E_B}(E_X)$, while (2.26) gives

$$q_{E_B}(A) = q_{E_B}\left(\bigcup_{X \geq B} E_X\right) = \bigcup_{X \geq B} q_{E_B}(E_X).$$

Consequently every affine centre is killed on a final segment of the endpoint filtration.

Finally, if Q is a prime-power step, then

$$E_{Q-1} = F_{<Q}, \quad E_Q = F_Q. \tag{2.28}$$

Indeed a prime-power rank is $< Q$ exactly when it is at most $Q - 1$. Thus the local simple quotient (2.24) is literally the associated quotient of the global endpoint filtration at Q .

3. FOUR PRIMITIVE INPUTS

The proof has four primitive inputs: a reciprocal-neutral count switch, comparable prime supply, a restricted fixed-cardinality sum theorem, and a finite independent-transversal quota. The first three are proved here or in the appendices in the exact form used below; the fourth is a direct finite-row consequence of Haxell's independent-transversal theorem.

3.1. Reciprocal-neutral component switches. Put

$$P(n) = \frac{1}{n} + \frac{1}{n+1}.$$

For every integer $a \geq 1$ one has the exact identity

$$P(a) + P(a^2 + 3a + 1) = P(a + 2) + P\left(\frac{a(a+3)}{2}\right) + P(a(a+3)). \quad (3.1)$$

Indeed, if $C = a(a+3)$, then $(a+1)(a+2) = C+2$ and

$$P(a) - P(a+2) = \frac{2(2C+3)}{C(C+2)},$$

while

$$P(C) - P(C+1) + P(C/2) = \frac{2(2C+3)}{C(C+2)}.$$

For $a \geq 3$ the two sides of (3.1) belong to C , with respectively two and three connected components. Indeed $C = a(a+3)$ is even, and on the right the successive starts $a+2, C/2, C$ are separated because $C/2 - a - 4 = (a^2 + a - 8)/2 \geq 2$ and $C/2 - 2 \geq 7$; the two starts on the left are farther apart. Their reciprocal values agree, while their component counts differ by one. As $a \rightarrow \infty$, their supports move arbitrarily far to the right and their common reciprocal value tends to zero.

Lemma 3.1 (Remote neutral Boolean family). *Let $\mathcal{O}$ be a finite family of interval components, let $\varepsilon > 0$, and let $L \geq 0$. There exist L mutually compatible copies of (3.1), all compatible with every component in $\mathcal{O}$, such that the sum of their common reciprocal values is less than ε .*

The resulting Boolean family has a common reciprocal value and hence a common residue translation, while its possible increments in component count are exactly

$$0, 1, \dots, L.$$

For each coordinate, at most five component intervals occur across its two alternatives.

Proof. Define the coordinates recursively. At stage i , let F_i be the finite union of the components already constructed together with $\mathcal{O}$, and let a_i be the least integer $a \geq 3$ for which both alternatives in (3.1) lie strictly to the right of F_i by more than one vertex and have common reciprocal value below $\varepsilon/(L+1)$. Such integers exist because the supports tend to infinity and the common reciprocal value tends to zero. This recursion gives mutual compatibility and total reciprocal value below ε . The two alternatives contain respectively two and three

components, so at most five component intervals occur in one coordinate. The Hamming-weight map

$$\{0, 1\}^L \rightarrow \{0, \dots, L\}$$

realizes every component-count increment. $\square$

3.2. Comparable prime bands. We use the elementary Chebyshev consequence proved in Appendix A.

Proposition 3.2 (Comparable prime supply). *There are an integer $\Lambda \geq 2$ and constants $c_\Pi, C_\Pi > 0$ such that, for every sufficiently large integer X ,*

$$c_\Pi \frac{X}{\log X} \leq \#\{p : X < p \leq \Lambda X\} \leq C_\Pi \frac{X}{\log X}.$$

3.3. Restricted sums in a prime cyclic group. The additive-combinatorial input is the Dias da Silva–Hamidoune theorem [6], proved in the required form as Theorem B.1 in Appendix B. Thus if S is cyclic of prime order p , $A \subseteq S$ has cardinality r , and

$$h(r - h) + 1 \geq p,$$

then every element of S is the sum of an h -element subset of A .

3.4. Independent thinning with a uniform quota. We use the 2Δ form of Haxell’s independent-transversal theorem; see [10, Theorem 2].

Proposition 3.3 (Independent quota lemma). *For a finite graph G whose vertices are partitioned into finite classes R_i , and for an integer $L \geq 1$, put*

$$\text{IQ}_L(G; (R_i)) = \left\{ P \subseteq V(G) : \begin{array}{l} P \text{ is independent, and} \\ |P \cap R_i| \geq \lfloor |R_i|/L \rfloor \text{ for every } i \end{array} \right\}.$$

For every integer $\Delta \geq 0$ set

$$L_P(\Delta) = \max\{1, 2\Delta\}, \quad \rho_P(\Delta) = \frac{1}{2L_P(\Delta)}. \quad (3.2)$$

Whenever G has maximum degree at most Δ ,

$$\text{IQ}_{L_P(\Delta)}(G; (R_i)) \neq \emptyset. \quad (3.3)$$

Moreover every $P \in \text{IQ}_{L_P(\Delta)}(G; (R_i))$ satisfies, whenever $|R_i| \geq L_P(\Delta)$,

$$|P \cap R_i| \geq \rho_P(\Delta)|R_i|.$$

Proof. For $\Delta = 0$ the full vertex set belongs to $\text{IQ}_1(G; (R_i))$. Suppose $\Delta > 0$. Split each class into as many full blocks of size 2Δ as possible and discard the remainder, and let G' be the subgraph induced by the union of the retained blocks. Its maximum degree is still at most Δ . Haxell’s theorem applied in G' to the partition by these blocks implies that the set of independent transversals of the block partition is nonempty. Every such transversal belongs to $\text{IQ}_{2\Delta}(G; (R_i))$. Finally, for $n \geq L_P(\Delta)$,

$$\left\lfloor \frac{n}{L_P(\Delta)} \right\rfloor \geq \frac{n}{2L_P(\Delta)} = \rho_P(\Delta)n.$$

$\square$

4. SIGNED-INVERSE FAMILIES AT PRIME-POWER STEPS

For every prime power $Q = p^e$, this section defines a finite family $\mathcal{A}_Q \subset \mathcal{C}$ and its simple-quotient map $\phi_Q : \mathcal{A}_Q \rightarrow S_Q^\times$. The construction is allowed to be empty at small Q ; the density, fibre, incompatibility, support and reciprocal-mass estimates used later are eventual. A single onset Q_{si} for all of those later capabilities is fixed at the end of the section.

Let Λ be the integer supplied by Proposition 3.2, and put

$$B_{\text{car}} = \Lambda + 1.$$

Let $Q = p^e$ and define

$$X_Q = \left\lfloor \frac{Q}{B_{\text{car}}} \right\rfloor. \quad (4.1)$$

Define the carrier set

$$\mathcal{B}_Q = \{b \text{ prime} : X_Q < b \leq \Lambda X_Q, (b, Q) = 1\}. \quad (4.2)$$

Writing

$$Q = B_{\text{car}} X_Q + r, \quad 0 \leq r < B_{\text{car}},$$

shows that every $b \in \mathcal{B}_Q$ satisfies

$$B_{\text{car}} b > Q, \quad b \leq \Lambda X_Q < B_{\text{car}} X_Q \leq Q. \quad (4.3)$$

For $X_Q \geq 1$ one also has

$$B_{\text{car}} X_Q \leq Q < 2B_{\text{car}} X_Q,$$

so $X_Q \asymp Q$. Once X_Q lies beyond the onset in Proposition 3.2, that proposition gives $\asymp Q/\log Q$ possible carriers; imposing $(b, Q) = 1$ removes at most the prime p .

For each $b \in \mathcal{B}_Q$, let $c_+ \in \{1, \dots, Q-1\}$ be the inverse of b modulo Q , put $c_- = Q - c_+$, and write

$$bc_+ = Qk_+ + 1, \quad bc_- = Qk_- - 1. \quad (4.4)$$

Then

$$1 \leq k_\pm < b < Q, \quad k_+ + k_- = b.$$

Since $p \nmid b$, the two coefficients cannot both be divisible by p . Define a sign deterministically: if $e = 1$, use the sign with larger coefficient, breaking a tie in favour of $+$; if $e \geq 2$, use $+$ unless $p \mid k_+$, and use $-$ otherwise. Let $k = k(Q, b)$ be the coefficient determined by this rule, and define

$$I(Q, b) = \begin{cases} [Qk, Qk+1], & +, \\ [Qk-1, Qk], & -. \end{cases} \quad (4.5)$$

If $Q = p$ is prime, this rule gives

$$k \geq b/2 > X_Q/2 \gg p. \quad (4.6)$$

4.1. Transversality and deterministic deduplication. The distinguished denominator Qk has largest prime-power stage exactly Q : the coefficient k is a p -unit, and every prime-power factor contributed by k has rank below Q because $k < Q$.

Lemma 4.1 (Carrier transversality and coefficient multiplicity). *For all sufficiently large prime powers $Q = p^e$ the following hold.*

- (1) *All but $O_{B_{\text{car}}}(1)$ carrier primes have companion denominator with no prime-power factor of rank at least Q .*
- (2) *A fixed coefficient k is produced by at most four carrier primes.*

Proof. For the companion denominator $bc_{\pm} = Qk \pm 1$, both factors b and $c_{\pm}$ are below Q . Suppose that a prime power $\ell^r \geq Q$ divides $bc_{\pm}$. If $\ell \neq b$, then the primality of b forces $\ell^r \mid c_{\pm}$, contradicting $c_{\pm} < Q$. Hence $\ell = b$. Since also $b < Q$, one has $r \geq 2$, and therefore $b \mid c_{\pm}$. Write $c_{\pm} = \ell_0 b$. Since $c_{\pm} < Q < B_{\text{car}}b$, one has $1 \leq \ell_0 < B_{\text{car}}$, and (4.4) gives

$$\ell_0 b^2 \equiv \pm 1 \pmod{Q}. \quad (4.7)$$

For any $(\ell_0, \pm)$, if $p \mid \ell_0$, the congruence has no solution because its right-hand side is a unit modulo p . Otherwise it reduces to a unit square congruence modulo p^e , which has at most two roots for odd p and at most four roots for $p = 2$. For the latter assertion, when $e \geq 3$ the quotient of any two unit roots is a solution of $u^2 \equiv 1 \pmod{2^e}$, of which there are four; the cases $e \leq 2$ are immediate. There are only $2(B_{\text{car}} - 1)$ choices of $(\ell_0, \pm)$, proving the first assertion.

For any coefficient k , every carrier producing it divides one of $Qk - 1$ and $Qk + 1$. Since $1 \leq k < Q$, both integers are smaller than Q^2 . Every carrier satisfies $b > Q/B_{\text{car}}$ by (4.1) and (4.2). If $Q > B_{\text{car}}^3$, a number below Q^2 cannot contain three distinct carrier primes, because their product would exceed $Q^3/B_{\text{car}}^3 > Q^2$. Thus each of $Qk - 1$ and $Qk + 1$ contains at most two possible carriers, giving multiplicity at most four. $\square$

For every Q , discard exactly those carriers whose companion denominator has a prime-power factor of rank at least Q . Lemma 4.1 shows that this deletes only $O_{B_{\text{car}}}(1)$ carriers once Q is sufficiently large. Different remaining carriers can still produce the same coefficient k ; keep, for each coefficient, the occurrence with least carrier. The eventual multiplicity bound in the lemma shows that, for sufficiently large Q , the resulting coefficient set has cardinality

$$\asymp Q/\log Q.$$

If $e \geq 2$, order the remaining coefficients $k_1 < \dots < k_N$ and retain the upper half. Since $k_i \geq i$, every retained coefficient then satisfies

$$k \gg Q/\log Q. \quad (4.8)$$

If $Q = p$ is prime, retain all coefficients. Let $\mathcal{A}_Q$ be the resulting family of intervals (4.5).

Lemma 4.2 (Top-stage recovery). *For every $I \in \mathcal{A}_Q$, exactly one of its two denominators has largest prime-power stage Q , namely the distinguished denominator Qk . Consequently I determines its stage Q and coefficient k ; in particular the families $\mathcal{A}_Q$ for distinct Q are disjoint.*

Proof. The distinguished denominator has top stage Q because k is a p -unit and $k < Q$. The transversality deletion removes every carrier for which the companion denominator has a prime-power factor of rank at least Q . Hence the companion has strictly smaller top stage. $\square$

4.2. The simple-quotient value. Modulo $F_{<Q}$, the companion reciprocal has only smaller prime-power stages, so the class contributed by $I(Q, b)$ is the class of $1/(Qk)$. Let $u \in \{1, \dots, p-1\}$ be the unique inverse of k modulo p , and write

$$uk = 1 + pt.$$

Then

$$\frac{u}{Q} - \frac{1}{Qk} = \frac{t}{p^{e-1}k}. \quad (4.9)$$

Every prime-power factor of the denominator on the right has rank below Q : the p -part is at most p^{e-1} , and every factor coming from k is below Q . Hence the difference belongs to $F_{<Q}$. In particular,

$$\overline{W}(I(Q, b)) \in F_Q, \quad (4.10)$$

and, relative to the distinguished generator represented by $1/Q$, its image in $S_Q = F_Q/F_{<Q}$ is

$$\phi_Q(I(Q, b)) = k^{-1} \pmod{p}. \quad (4.11)$$

Thus the simple-quotient map

$$\phi_Q : \mathcal{A}_Q \longrightarrow S_Q^\times$$

is well defined. By Lemma 4.2, the distinguished-point map $I(Q, b) \mapsto Qk$ is globally injective.

For an integer interval $J = [u, v] \subset \mathbb{N}_{>0}$ define its *exclusion envelope*

$$\text{Env}(J) = \{n \in \mathbb{N}_{>0} : \text{dist}(n, J) \leq 1\}.$$

Thus another interval component is incompatible with J exactly when its support meets $\text{Env}(J)$. A member of some $\mathcal{A}_Q$ is said to meet an exclusion envelope if its support intersects that envelope.

Proposition 4.3 (Uniform estimates for the signed-inverse families). *The deterministic families $\mathcal{A}_Q$ satisfy the following properties.*

(1) *Their normalized density has a positive lower limit:*

$$\kappa_\infty := \liminf_{\substack{Q \rightarrow \infty \\ Q \text{ prime power}}} \frac{|\mathcal{A}_Q| \log Q}{Q} > 0. \quad (4.12)$$

(2) *For every sufficiently large $Q = p^e$ and every $s \in S_Q^\times$,*

$$|\phi_Q^{-1}(s)| \leq Q/p. \quad (4.13)$$

(3) *The incompatibility graph on $\bigcup_Q \mathcal{A}_Q$ has finite maximum degree. More generally, for every fixed $L \geq 1$ there is C_L such that an exclusion envelope of cardinality at most L meets at most C_L members of the union.*

(4) The normalized lower support edge has positive lower limit:

$$c_\infty := \liminf_{\substack{Q \rightarrow \infty \\ Q \text{ prime power}}} \left(\min_{\substack{I \in \mathcal{A}_Q \\ z \in I}} \frac{z \log Q}{Q^2} \right) > 0, \quad (4.14)$$

where the minimum is taken only for the eventually nonempty families. Every support point also satisfies $z \leq 2Q^2$ for all sufficiently large Q , and if Q is prime then $z \gg Q^2$.

(5) Define

$$\omega_Q = \max \left(\{0\} \cup \left\{ W \left(\bigcup_{I \in O} I \right) : O \subseteq \mathcal{A}_Q, |O| \leq p, \bigcup_{I \in O} I \in \mathcal{C} \right\} \right). \quad (4.15)$$

Then

$$\sum_Q \omega_Q < \infty. \quad (4.16)$$

Proof. The positive lower density follows from the carrier count, the bounded exceptional set, bounded coefficient multiplicity and, for $e \geq 2$, the upper-half thinning. For (4.13), equality of simple-quotient values implies equality of k modulo p . A fixed nonzero residue class modulo p occurs exactly Q/p times among $1 \leq k < Q = p^e$, and the deterministic deduplication retains at most one interval for each coefficient.

For incompatibility, each interval occupies two adjacent integers around its globally injective distinguished point Qk . If an exclusion envelope has cardinality at most L , every member meeting it has distinguished point in a fixed one-step enlargement of the envelope. This enlargement contains $O(L)$ integer sites and global injectivity leaves at most one retained interval per site. In particular the global incompatibility graph has bounded integer degree.

The positive lower support limit follows from $z = Qk + O(1)$ together with (4.8); the prime lower bound follows from (4.6). The upper bound $z \leq 2Q^2$ follows from $k < Q$.

For the quantity (4.15), a prime family has $O(p/\log p)$ intervals, each of reciprocal mass $O(p^{-2})$, so

$$\omega_p = O\left(\frac{1}{p \log p}\right).$$

For $Q = p^e$, $e \geq 2$, each interval has mass $O(\log Q/Q^2)$, so

$$\omega_{p^e} = O\left(\frac{p \log Q}{Q^2}\right) = O\left(\frac{e \log p}{p^{2e-1}}\right).$$

For the prime terms, let $X_j = \Lambda^j X_0$, with X_0 beyond the onset of Proposition 3.2. The contribution of primes in $X_j < p \leq \Lambda X_j$ is at most

$$O\left(\frac{X_j}{\log X_j}\right) O\left(\frac{1}{X_j \log X_j}\right) = O\left(\frac{1}{(\log X_j)^2}\right) = O(j^{-2}),$$

so the prime terms in the tail are summable. The composite tail is bounded by a constant multiple of $\sum_p p^{-2}$ after summing over $e \geq 2$. The finitely many remaining prime-power stages contribute a finite amount, hence (4.16) holds. $\square$

Set the positive constants

$$\kappa = \frac{\kappa_\infty}{2}, \quad c = \frac{1}{2} \min\{1, c_\infty\}. \quad (4.17)$$

Here the lower limits are allowed to take values in the extended real line; the truncation ensures that c is finite even if $c_\infty = +\infty$. The carrier upper bound makes κ_∞ finite. By the definitions of the lower limits, for every sufficiently large prime power Q one has

$$|\mathcal{A}_Q| \geq \kappa \frac{Q}{\log Q} \quad (4.18)$$

and every support point satisfies

$$c \frac{Q^2}{\log Q} \leq z \leq 2Q^2. \quad (4.19)$$

Choose once and for all an integer Q_{si} so large that, for every prime power $Q \geq Q_{\text{si}}$, the carrier-transversality and coefficient-multiplicity bounds, the fibre bound (4.13), the family-size estimate (4.18), and the support bounds (4.19) (including the prime lower bound used above) all hold. The objects $\mathcal{A}_Q$, ϕ_Q and ω_Q remain defined for every prime power Q ; the finitely many stages below Q_{si} are simply outside the later transport regime. Every transport onset introduced below is taken to be at least Q_{si} .

Let G_∞ be the incompatibility graph on $\bigcup_Q \mathcal{A}_Q$. Proposition 4.3 shows that the bounded subset

$$\{\deg_{G_\infty}(v) : v \in V(G_\infty)\} \subseteq \mathbb{N}$$

has a maximum; define

$$\Delta_* = \max_{v \in V(G_\infty)} \deg_{G_\infty}(v). \quad (4.20)$$

Abbreviate

$$L_P = L_P(\Delta_*), \quad \rho_P = \rho_P(\Delta_*), \quad (4.21)$$

with the functions of (3.2). These constants will be used in the finite compatibility step.

In particular, for every $\varepsilon > 0$ there is Q_1 such that every finite set of prime-power steps of rank at least Q_1 has total reciprocal-mass bound below ε .

5. A WIDE AFFINE INITIAL COVER

This section constructs the admissible initial covers. Prime-power bridges are pasted by the affine pullback of Section 2; a Hamming-weight pullback then turns the bounded count spread into an interval of prescribed subscale width.

Put

$$L_n = \text{lcm}(1, \dots, n).$$

At a prime-power jump $n = p^e$, write

$$D = L_{n-1}, \quad L_n = pD.$$

By Lemma 2.8, the quotient step has order p .

5.1. A uniform bridge across one simple quotient. Let k be the least positive integer satisfying

$$T_k = \frac{k(k+1)}{2} \geq p-1,$$

and put $R_p = \{1, \dots, k\}$. The subset sums of R_p fill $[0, T_k]$, and hence reduce surjectively modulo p .

Minimality gives $k \leq p-1$, so every $r \in R_p$ divides D . Set

$$a_r = \frac{pD}{r}$$

and use the two alternatives

$$[a_r + 1, a_r + 2] \longleftrightarrow [a_r, a_r + 2]. \quad (5.1)$$

Their reciprocal difference is exactly $r/(pD)$. Thus the Boolean branch object on R_p maps surjectively to

$$L_n^{-1}\mathbb{Z}/L_{n-1}^{-1}\mathbb{Z} \cong \mathbb{Z}/p\mathbb{Z}.$$

Every branch contains exactly k connected components, so the bridge changes the residue class without introducing any branch-dependent component-count change.

Minimality also gives $k^2 = O(p)$ and $T_k = O(p)$. The total reciprocal mass of the bridge is therefore $O(D^{-1})$. If $r < s \leq k$, then

$$a_r - a_s = \frac{pD(s-r)}{rs} \geq \frac{pD}{k^2}. \quad (5.2)$$

For all sufficiently large jumps the right-hand side is at least 4, so every alternative at one bridge base is disjoint and nonadjacent from every alternative at another base. Lemma C.1 supplies the finite initialization before this uniform separation regime begins.

At each nontrivial least-common-multiple jump the next lower modulus is the present upper modulus pD with $p \geq 2$, so successive nontrivial lower moduli at least double. Hence the total reciprocal mass of the future bridge sequence is finite. Together with the finite initialization this leaves a fixed positive margin below 2.

At the jump $n = p^e$, one has $E_{n-1} = H_D$ and $E_n = H_{pD}$. Let $\mathcal{B}_n$ be the finite family of all bridge labels obtained from the Boolean alternatives in (5.1). Any two branches differ by a sum of classes $r/(pD) \in E_n$, so the composite

$$\mathcal{B}_n \xrightarrow{\overline{W}} A \xrightarrow{q_{E_n}} A/E_n$$

is constant; denote its value by β_n . The preceding subset-sum calculation is exactly the regular epimorphism

$$q_{E_{n-1}} \overline{W} : \mathcal{B}_n \twoheadrightarrow \text{Fib}_{\pi_{E_{n-1}, E_n}}(\beta_n). \quad (5.3)$$

Every branch has the same component count k .

Thus Corollary 2.6, with $H = E_{n-1}$ and $K = E_n$, gives: an incoming regular epimorphism onto an affine E_{n-1} -fibre pastes with (5.3) to a regular epimorphism onto the corresponding affine E_n -fibre, while the component-count spread is unchanged.

5.2. Bounded-spread affine cover. Use the finite initialization of Lemma C.1 and paste it with one bridge object at every subsequent prime-power jump up to a cutoff B . Repeated application of Corollary 2.6 produces a finite parameter set $\tilde{\mathcal{B}}_B$, a label map

$$\lambda_B : \tilde{\mathcal{B}}_B \longrightarrow C,$$

a centre $\alpha_B \in A/E_B$, and a regular epimorphism

$$\overline{W}\lambda_B : \tilde{\mathcal{B}}_B \twoheadrightarrow \text{Fib}_{E_B}(\alpha_B). \quad (5.4)$$

The parameter set is retained here because its points, not merely their label image in C , carry the covering map used in the next pullback. Since every bridge adds the same number of components to every label, the difference between the largest and smallest component counts remains equal to the finite initial spread. Thus there is a constant G_0 and integers $m_B \leq M_B$ such that

$$\nu(\lambda_B(b)) \in [m_B, M_B] \quad (b \in \tilde{\mathcal{B}}_B), \quad M_B - m_B = G_0. \quad (5.5)$$

The explicit integers of Appendix C are used only to establish this finite initial cover.

5.3. Sparse interaction with later prime-power families. Let $Q > B$ be a later prime-power step. Proposition 4.3 places the support of $\mathcal{A}_Q$ inside

$$\left[c \frac{Q^2}{\log Q}, 2Q^2 \right] \quad (5.6)$$

up to a fixed endpoint enlargement.

Lemma 5.1 (Sparse bridge interaction). *Uniformly for later prime-power steps $Q > B$, the fixed seed and the bridge stages used in the affine cover contribute only*

$$O(\sqrt{\log Q} \log \log Q) = o(Q/\log Q)$$

bounded-size exclusion envelopes capable of meeting $\mathcal{A}_Q$.

Proof. Suppose a bridge at a jump $n = p^e$, with lower modulus $D = L_{n-1}$, can meet this interval. Since $1 \leq r \leq k \leq p-1$, every bridge base satisfies

$$D < a_r = \frac{pD}{r} \leq pD.$$

Intersection with the upper endpoint of (5.6) therefore gives

$$D = O(Q^2).$$

The primorial through $n-1$ divides D , so the Chebyshev estimate of Appendix A gives

$$n, p = O(\log Q).$$

Intersection with the lower endpoint of (5.6), together with $a_r \leq pD$, gives

$$D \gg \frac{Q^2}{p \log Q} \gg \frac{Q^2}{(\log Q)^2}.$$

Thus there are fixed constants $c_1, c_2 > 0$ such that every relevant lower modulus satisfies

$$c_1 \frac{Q^2}{(\log Q)^2} \leq D \leq c_2 Q^2.$$

Consequently the ratio between the largest and smallest relevant lower moduli is $O((\log Q)^2)$. If $D_1 < \dots < D_t$ are the relevant lower moduli in chronological order, every nontrivial least-common-multiple jump gives $D_{j+1} \geq 2D_j$. Hence

$$2^{t-1} \leq O((\log Q)^2),$$

so $t = O(\log \log Q)$. Therefore only $O(\log \log Q)$ bridge stages can meet the support window. Each contains

$$k = O(\sqrt{p}) = O(\sqrt{\log Q})$$

bounded-size exclusion envelopes. The total number of bridge envelopes capable of meeting $\mathcal{A}_Q$ is therefore

$$O(\sqrt{\log Q} \log \log Q) = o(Q/\log Q). \quad (5.7)$$

The fixed finite seed contributes only $O(1)$ further bounded-size exclusion envelopes. This proves the lemma. $\square$

By Proposition 4.3, the envelopes in Lemma 5.1 are incompatible with only $o(Q/\log Q)$ members of $\mathcal{A}_Q$.

5.4. The neutral Hamming pullback. Let

$$\mathfrak{L} = \left\{ L : \mathbb{N}_{>0} \rightarrow \mathbb{N} : \lim_{B \rightarrow \infty} \frac{L(B) \log B}{B} = 0 \right\}. \quad (5.8)$$

For $L \in \mathfrak{L}$ put

$$N_B = G_0 + L(B), \quad J_B = \{0, \dots, L(B)\}.$$

For an arbitrary N_B -coordinate neutral Boolean family $\mathcal{N}$ as in Lemma 3.1, let

$$\Psi_{\mathcal{N}} : \{0, 1\}^{N_B} \rightarrow C$$

be its configuration map. It has constant reciprocal value and component count

$$\nu(\Psi_{\mathcal{N}}(v)) = 2N_B + |v|.$$

Define

$$\chi_B : \tilde{\mathcal{B}}_B \times J_B \rightarrow \{0, \dots, N_B\}, \quad \chi_B(b, j) = M_B - \nu(\lambda_B(b)) + j.$$

By (5.5), this is well defined. The Hamming-weight map is a regular epimorphism, and the cartesian square

$$\begin{array}{ccc} \mathcal{P}_B(\mathcal{N}) & \longrightarrow & \{0, 1\}^{N_B} \\ p_B \downarrow & & \downarrow |\cdot| \\ \tilde{\mathcal{B}}_B \times J_B & \xrightarrow{\chi_B} & \{0, \dots, N_B\} \end{array} \quad (5.9)$$

therefore makes p_B a regular epimorphism. The compatible-union map

$$u_B : \mathcal{P}_B(\mathcal{N}) \rightarrow C, \quad (b, j, v) \mapsto \lambda_B(b) \cup \Psi_{\mathcal{N}}(v)$$

has component count

$$\nu(u_B(b, j, v)) = 2N_B + M_B + j.$$

Consequently the count coordinate covers the interval

$$I_B = [2N_B + M_B, 2N_B + M_B + L(B)], \quad (5.10)$$

whose width is exactly $L(B)$.

Let $\tau_{\mathcal{N}} = \overline{W}(\Psi_{\mathcal{N}}(v))$, independent of v , and put

$$\gamma_B = \alpha_B + q_{E_B}(\tau_{\mathcal{N}}) \in A/E_B. \quad (5.11)$$

Define

$$\begin{aligned} \Theta_{B,\mathcal{N}} : \mathcal{P}_B(\mathcal{N}) &\longrightarrow \text{Fib}_{E_B}(\gamma_B) \times I_B, \\ (b, j, v) &\longmapsto (\overline{W}(u_B(b, j, v)), 2N_B + M_B + j). \end{aligned} \quad (5.12)$$

Let $t_{\tau} : \text{Fib}_{E_B}(\alpha_B) \rightarrow \text{Fib}_{E_B}(\gamma_B)$ be translation by $\tau_{\mathcal{N}}$ and let $s_B : J_B \rightarrow I_B$ send j to $2N_B + M_B + j$. Both maps are isomorphisms. The defining count identity gives the factorization

$$\Theta_{B,\mathcal{N}} = (t_{\tau} \times s_B) \circ ((\overline{W}\lambda_B) \times 1_{J_B}) \circ p_B.$$

The three maps are respectively an isomorphism, a product of regular epimorphisms, and a pullback of the Hamming-weight regular epimorphism. Hence $\Theta_{B,\mathcal{N}}$ is a regular epimorphism. With source constantly 0_M and label map u_B , this defines the covering labelled correspondence

$$R_{B,\mathcal{N}} : \{0_M\} \rightsquigarrow \text{Fib}_{E_B}(\gamma_B) \times I_B. \quad (5.13)$$

5.5. Uniform mass and compatibility parameters. Appendix C gives the uniform bound $W < 7/4$ for every initialized branch together with the entire subsequent bridge sequence. Set

$$\eta_0 = \frac{1}{4}. \quad (5.14)$$

Let

$$0 < \eta < \eta_0, \quad 0 < \sigma < 1$$

be arbitrary. Let $\mathfrak{N}_B(\eta)$ denote the set of all N_B -coordinate neutral Boolean families compatible with the seed and bridge components and having total common reciprocal value below $\eta_0 - \eta$. Lemma 3.1 says that $\mathfrak{N}_B(\eta)$ is nonempty. For every $\mathcal{N} \in \mathfrak{N}_B(\eta)$, every edge label of (5.13) has reciprocal value below $2 - \eta$.

For $\mathcal{N} \in \mathfrak{N}_B(\eta)$ let $\mathcal{P}_{B,\mathcal{N}}$ be the finite pool of connected interval components that occur in the seed, bridge alternatives through B , or neutral-coordinate alternatives. For every later prime-power stage put

$$\mathcal{A}_Q(B, \mathcal{N}) = \{I \in \mathcal{A}_Q : I \text{ is compatible with every } J \in \mathcal{P}_{B,\mathcal{N}}\}. \quad (5.15)$$

Every connected component in the pool has an exclusion envelope of cardinality at most 5. Let C_5 be the uniform envelope bound from Proposition 4.3. Put

$$\varepsilon_{\sigma} = \frac{(1 - \sigma)\kappa}{2} > 0.$$

By Lemma 5.1, after an absolute onset the seed and relevant bridges delete at most

$$\varepsilon_{\sigma} \frac{Q}{\log Q}$$

members of $\mathcal{A}_Q$. The neutral family contributes at most $5N_B$ component envelopes. Since $L(B) = o(B/\log B)$, for every fixed $L \in \mathfrak{Q}$ and $\sigma < 1$ there is $B_0(\sigma, L)$ such that

$$5C_5N_B \leq \varepsilon_{\sigma} \frac{B}{\log B}$$

for $B \geq B_0(\sigma, L)$. As $Q/\log Q$ is eventually increasing, the same bound holds with B replaced by every $Q \geq B$. Hence the number of deleted members is at most

$$(1 - \sigma)\kappa \frac{Q}{\log Q} \leq (1 - \sigma)|\mathcal{A}_Q|,$$

and therefore

$$|\mathcal{A}_Q(B, \mathcal{N})| \geq \sigma|\mathcal{A}_Q| \quad (5.16)$$

for every later prime-power step $Q > B$.

A prepared start at B for (η, σ, L) is the tuple

$$\xi = (R_{B, \mathcal{N}}, \gamma_B, I_B, \mathcal{P}_{B, \mathcal{N}})$$

produced by a parameter $\mathcal{N} \in \mathfrak{N}_B(\eta)$ as above. Let $\mathfrak{W}_B(\eta, \sigma, L)$ denote the set of all prepared starts. For

$$\xi \in \mathfrak{W}_B(\eta, \sigma, L)$$

write $R_B^\xi, \gamma_B^\xi, I_B^\xi$ and $\mathcal{P}_B^\xi$ for its four components, and put

$$\mathcal{A}_Q(\xi) = \{I \in \mathcal{A}_Q : I \text{ is compatible with every } J \in \mathcal{P}_B^\xi\}.$$

Proposition 5.2 (Wide affine initial cover). *With η_0 defined by (5.14), for every*

$$0 < \eta < \eta_0, \quad 0 < \sigma < 1, \quad L \in \mathfrak{L}$$

there is $B_0(\eta, \sigma, L)$ for which

$$\mathfrak{W}_B(\eta, \sigma, L) \neq \emptyset \quad (B \geq B_0(\eta, \sigma, L)). \quad (5.17)$$

Every $\xi \in \mathfrak{W}_B(\eta, \sigma, L)$ contains a covering labelled correspondence

$$R_B^\xi : \{0_M\} \rightsquigarrow \text{Fib}_{E_B}(\gamma_B^\xi) \times I_B^\xi,$$

where the width of I_B^ξ is $L(B)$, every edge label has reciprocal value below $2 - \eta$, and every later prime-power family satisfies

$$|\mathcal{A}_Q(\xi)| \geq \sigma|\mathcal{A}_Q|.$$

The onset is chosen before the neutral family: the survival estimate holds for every $\mathcal{N} \in \mathfrak{N}_B(\eta)$ and every $Q > B$. This order matters because the supports of the neutral switches may lie arbitrarily far beyond B .

6. UNIFORM TRANSFER ON THE SIMPLE QUOTIENTS

Using the local arithmetic estimates of Section 4, this section proves a uniform transfer theorem across each simple quotient. For every sufficiently dense restriction $E \subseteq \mathcal{A}_Q$, the required incoming width is bounded by $\delta_\rho(Q)$, and prime stages additionally contribute a positive gain $\mu_\rho(q)$.

Let S be a cyclic group of prime order p , let E be a finite set, and let

$$f : E \longrightarrow S^\times.$$

Define the component-count graded subset-sum support

$$\Gamma_f = \left\{ \left(\sum_{x \in O} f(x), |O| \right) : O \subseteq E, |O| \leq p \right\} \subseteq S \times \mathbb{N}. \quad (6.1)$$

At this stage Γ_f is an abstract subset-sum support; the corresponding physical unions of interval components are constructed only after the finite compatibility

selection in Section 8. If $I \subset \mathbb{N}$ is an incoming interval of component counts, the induced target support is

$$\Gamma_f + (\{0\} \times I).$$

6.1. Large image and heavy fibres. Put

$$B = \text{im } f, \quad r = |B|.$$

For $0 \leq h \leq r$, the occurrence relation from h -element subsets of B to h distinct elements of E is surjective: each value fibre is nonempty, and distinct values require distinct occurrences. Hence Theorem B.1 applies directly.

Let

$$I_D(p, r) = \{h \in \{0, \dots, r\} : h(r - h) + 1 \geq p\}.$$

If this set is nonempty and $a = \min I_D(p, r)$, symmetry and unimodality give

$$I_D(p, r) = [a, r - a] \cap \mathbb{N},$$

and

$$a < \frac{2p}{r} + 1. \quad (6.2)$$

Indeed, minimality gives $(a - 1)(r - a + 1) < p - 1$, while $a \leq r/2$; hence $(a - 1)r/2 < p - 1$. The large-image case therefore gives

$$S \times [A_0 + a, B_0 + r - a] \subseteq \Gamma_f + (\{0\} \times [A_0, B_0]). \quad (6.3)$$

The complementary mechanism uses a heavy fibre. If some $c \in S^\times$ has at least $p - 1$ preimages, then for every $0 \leq j < p$ the full family of j -element subsets of $f^{-1}(c)$ gives

$$(jc, j) \in \Gamma_f.$$

Since $j \mapsto jc$ bijects $\{0, \dots, p - 1\}$ with S , one obtains

$$S \times [A_0 + p - 1, B_0] \subseteq \Gamma_f + (\{0\} \times [A_0, B_0]) \quad (6.4)$$

whenever the incoming interval has width at least $p - 1$.

6.2. Uniformity under dense restriction. Let κ be the density constant in (4.18), and let L_P, ρ_P be the packing constants in (4.21). Let

$$0 < \sigma < 1$$

be arbitrary, and let ρ be an arbitrary parameter satisfying

$$0 < \rho < \kappa \sigma \rho_P. \quad (6.5)$$

All estimates below are uniform for each such fixed pair (σ, ρ) ; later sections use the same σ in Proposition 5.2. For every sufficiently large prime power $Q = p^e$, define

$$\mathcal{D}_\rho(Q) = \left\{ E \subseteq \mathcal{A}_Q : |E| \geq \rho \frac{Q}{\log Q} \right\}. \quad (6.6)$$

For large Q this collection is nonempty, since (4.18) and (6.5) place $\mathcal{A}_Q$ itself in $\mathcal{D}_\rho(Q)$.

Let $E \in \mathcal{D}_\rho(Q)$ be arbitrary and restrict ϕ_Q to E . If

$$m_E = \max_s |\phi_Q^{-1}(s) \cap E|, \quad r = |\text{im}(\phi_Q|_E)|,$$

then (4.13) gives $m_E \leq Q/p$, and therefore

$$r \gg_\rho \frac{p}{\log Q}. \quad (6.7)$$

If $I_D(p, r)$ is nonempty, (6.2) gives

$$a(E) = O_\rho(\log Q). \quad (6.8)$$

If $I_D(p, r)$ is empty, then $r^2 < 4p$, and (6.7) yields

$$p = O_\rho((\log Q)^2). \quad (6.9)$$

Moreover some nonzero simple-quotient value has at least

$$\frac{|E|}{r} \gg_\rho \frac{Q}{\sqrt{p} \log Q} \quad (6.10)$$

preimages. If $Q = p^e$ with $e \geq 2$, this is eventually larger than $p - 1$ uniformly in p and e , because after division by p the relevant ratio is

$$\frac{Q}{p^{3/2} \log Q} = \frac{p^{e-3/2}}{e \log p}. \quad (6.11)$$

The required uniformity can be made explicit.

Lemma 6.1 (Uniform heavy-fibre threshold). *For every fixed $\rho > 0$ there is $Q_{\text{hf}}(\rho)$ such that, whenever $Q = p^e \geq Q_{\text{hf}}(\rho)$ is composite and $E \in \mathcal{D}_\rho(Q)$ lies in the heavy-fibre case, some nonzero simple-quotient value has at least $p - 1$ preimages in E .*

Proof. Since $e \geq 2$, one has $p \leq Q^{1/2}$. Consequently

$$\frac{Q}{p^{3/2} \log Q} \geq \frac{Q^{1/4}}{\log Q} \rightarrow \infty.$$

Together with (6.10), this gives a single threshold, independent of p and e , beyond which the heavy fibre has size at least $p - 1$. $\square$

Thus (6.4) applies to every sufficiently large composite prime power, with required incoming width

$$p - 1 = O_\rho((\log Q)^2).$$

If $Q = q$ is prime, (4.13) has capacity one, so $\phi_q|_E$ is injective and

$$r = |E| \geq \rho q / \log q.$$

The alternative $I_D(q, r) = \emptyset$ would imply $r^2 < 4q$, contradicting

$$r^2 \geq \rho^2 \frac{q^2}{(\log q)^2} > 4q$$

for sufficiently large q . Hence the large-image case is forced at prime steps and

$$r \gg_\rho q / \log q, \quad a(E) = O_\rho(\log q). \quad (6.12)$$

6.3. Uniform loss and prime gain. Fix ρ . Let $Q_{\text{tr}}(\rho) \geq Q_{\text{si}}$ be the least integer such that all preceding uniform statements hold for every prime power $Q \geq Q_{\text{tr}}(\rho)$.

For $E \in \mathcal{D}_\rho(Q)$ define

$$\delta(Q, E) = \begin{cases} a(E), & \text{in the large-image case,} \\ p - 1, & \text{in the heavy-fibre case.} \end{cases}$$

and then set

$$\delta_\rho(Q) = \max_{E \in \mathcal{D}_\rho(Q)} \delta(Q, E). \quad (6.13)$$

Thus

$$\delta_\rho(Q) = O_\rho((\log Q)^2) \quad (6.14)$$

for every sufficiently large prime power, while

$$\delta_\rho(q) = O_\rho(\log q) \quad (6.15)$$

when q is prime.

For a prime q and $E \in \mathcal{D}_\rho(q)$ put

$$r_E = |\text{im}(\phi_q|_E)|.$$

With $\delta_\rho(q)$ now fixed, define

$$\mu_\rho(q) = \min_{E \in \mathcal{D}_\rho(q)} (r_E - a(E) - \delta_\rho(q)). \quad (6.16)$$

By (6.12) and (6.15),

$$\mu_\rho(q) \gg_\rho q/\log q. \quad (6.17)$$

Replacing $Q_{\text{tr}}(\rho)$ by the least larger onset if necessary, we may and do require $\mu_\rho(q) > 0$ for every prime $q \geq Q_{\text{tr}}(\rho)$.

Theorem 6.2 (Uniform simple-quotient transfer). *Let $Q \geq Q_{\text{tr}}(\rho)$ and $E \in \mathcal{D}_\rho(Q)$. If Q is composite and the incoming component-count interval is $[A_0, B_0]$ of width at least $\delta_\rho(Q)$, then*

$$S_Q \times [A_0 + \delta_\rho(Q), B_0] \subseteq \Gamma_{\phi_Q|_E} + (\{0\} \times [A_0, B_0]).$$

If $Q = q$ is prime, then

$$S_q \times [A_0 + \delta_\rho(q), B_0 + \delta_\rho(q) + \mu_\rho(q)] \subseteq \Gamma_{\phi_q|_E} + (\{0\} \times [A_0, B_0]).$$

Proof. First suppose that $Q = p^e$ is composite. If E is in the large-image case, put $a = a(E)$ and $r = |\text{im}(\phi_Q|_E)|$. By definition $a \leq \delta_\rho(Q)$, while $a \leq r$. Hence

$$[A_0 + \delta_\rho(Q), B_0] \subseteq [A_0 + a, B_0 + r - a],$$

and the required inclusion follows from (6.3). If E is in the heavy-fibre case, then $p - 1 \leq \delta_\rho(Q)$ and therefore

$$[A_0 + \delta_\rho(Q), B_0] \subseteq [A_0 + p - 1, B_0],$$

so (6.4) applies. The width hypothesis ensures that the displayed target interval is nonempty in either case.

Now let $Q = q$ be prime. The large-image case is forced. Write $a = a(E)$ and $r_E = |\text{im}(\phi_q|_E)|$. By the definition of the uniform demand,

$$a \leq \delta_\rho(q),$$

and by the definition of the uniform prime gain,

$$\mu_\rho(q) \leq r_E - a - \delta_\rho(q).$$

Thus

$$[A_0 + \delta_\rho(q), B_0 + \delta_\rho(q) + \mu_\rho(q)] \subseteq [A_0 + a, B_0 + r_E - a].$$

The prime assertion follows again from (6.3). $\square$

7. CONTINUATION OF COMPONENT-COUNT INTERVALS

Fix

$$0 < \eta < \eta_0, \quad 0 < \sigma < 1, \quad 0 < \rho < \kappa \sigma \rho_P.$$

The triple (η, σ, ρ) is fixed before any onset, base or prepared start in this section. Every threshold introduced below depends only on this triple and on the fixed constants from the preceding sections; a prepared start is chosen only after its base satisfies those thresholds. Only the count intervals and the transported affine centre are retained in this section. Put

$$Q_{\text{tr}} = Q_{\text{tr}}(\rho), \quad \Phi(X) = \frac{X}{\log X}.$$

Theorem 6.2 supplies, at every prime-power step $Q \geq Q_{\text{tr}}$, an incoming-width demand $\delta_\rho(Q)$ and, when $Q = q$ is prime, a positive width gain $\mu_\rho(q)$.

For an integer interval $I = [A, B]$ write $w(I) = B - A$. At a prime q the scalar action is

$$[A, B] \mapsto [A + \delta_\rho(q), B + \delta_\rho(q) + \mu_\rho(q)], \quad (7.1)$$

while at a composite prime power Q it is

$$[A, B] \mapsto [A + \delta_\rho(Q), B]. \quad (7.2)$$

A step is available precisely when the incoming width is at least $\delta_\rho(Q)$; in that case the old and new integer intervals overlap.

Extend δ_ρ by 0 below Q_{tr} and define

$$\begin{aligned} \Delta_\rho(X) &= \max\left(\{0\} \cup \{\delta_\rho(Q) : Q \leq X, Q \text{ a prime power}\}\right), \\ \mathcal{L}_\rho(X) &= \sum_{\substack{Q_{\text{tr}} \leq p^e \leq X \\ e \geq 2}} \delta_\rho(p^e). \end{aligned} \quad (7.3)$$

By (6.14),

$$\Delta_\rho(X) = O_\rho((\log X)^2) = o(\Phi(X)). \quad (7.4)$$

The number of non-prime prime powers at most X is $O(\sqrt{X} \log X)$, hence

$$\mathcal{L}_\rho(X) = O_\rho(\sqrt{X} \log^3 X) = o(\Phi(X)). \quad (7.5)$$

7.1. Bounded-lag prime compensation. Let $\Lambda \geq 2$ be the multiplier in Proposition 3.2 and put

$$\Lambda_s = 4\Lambda. \quad (7.6)$$

For $Q \geq 1$ define the finite bounded-lag prime set

$$\text{Prev}_\rho(Q) = \{q : q \text{ prime}, Q_{\text{tr}} \leq q < Q \leq \Lambda_s q\}. \quad (7.7)$$

For all sufficiently large Q this set is nonempty: with $Y = \lfloor Q/(2\Lambda) \rfloor$, Proposition 3.2 supplies a prime in $(Y, \Lambda Y]$, and after a fixed onset it belongs to (7.7).

Whenever $\text{Prev}_\rho(Q) \neq \emptyset$ put

$$\underline{\mu}_\rho(Q) = \min_{q \in \text{Prev}_\rho(Q)} \mu_\rho(q). \quad (7.8)$$

Since $Q/\Lambda_s \leq q < Q$ throughout the bounded-lag prime set, (6.17) gives

$$\underline{\mu}_\rho(Q) \gg_{\rho, \Lambda_s} \frac{Q}{\log Q} = \Theta_{\rho, \Lambda_s}(\Phi(Q)). \quad (7.9)$$

Define the startup request

$$L_\rho(B) = \Delta_\rho(\Lambda_s B) + \mathcal{L}_\rho(\Lambda_s B). \quad (7.10)$$

Then

$$L_\rho(B) = o(B/\log B), \quad (7.11)$$

so $L_\rho \in \mathfrak{Q}$.

For a base B , let $\text{Cont}_\rho(B)$ denote the following purely scalar condition:

- (i) $B \geq Q_{\text{tr}}$;
- (ii) $\text{Prev}_\rho(Q) \neq \emptyset$ for every prime power $Q > \Lambda_s B$;
- (iii) for every such Q ,

$$\underline{\mu}_\rho(Q) - \mathcal{L}_\rho(Q) \geq \delta_\rho(Q). \quad (7.12)$$

Equations (7.9), (7.5) and (7.4) imply that $\text{Cont}_\rho(B)$ holds for every sufficiently large B . Put

$$\mathcal{B}_{\text{cont}}(\rho) = \{B \in \mathbb{N} : \text{Cont}_\rho(B)\}. \quad (7.13)$$

Thus $\mathcal{B}_{\text{cont}}(\rho)$ contains a final segment.

7.2. Prepared starts and the first-failure argument. By Proposition 5.2, for every sufficiently large B the prepared-start object

$$\mathfrak{B}_B(\eta, \sigma, L_\rho)$$

is nonempty. Fix a point ξ of this object and write

$$R_B^\xi : \{0_M\} \rightsquigarrow \text{Fib}_{E_B}(\gamma_B^\xi) \times I_B^\xi$$

for its covering correspondence. Its initial interval has width $L_\rho(B)$.

Process prime powers $Q > B$ in increasing order using (7.1) and (7.2). Whenever all earlier steps are available, let I_{Q-}^ξ denote the interval immediately before the Q -step and define its incoming width

$$w_\xi^-(Q) = w(I_{Q-}^\xi). \quad (7.14)$$

After the Q -step write I_Q^ξ for the resulting interval; between prime powers the interval is unchanged.

Lemma 7.1 (Continuation). *If $\text{Cont}_\rho(B)$ holds, then for every $\xi \in \mathfrak{B}_B(\eta, \sigma, L_\rho)$ all prime-power steps after B are available. More precisely,*

$$B < Q \leq \Lambda_s B \implies w_\xi^-(Q) \geq \delta_\rho(Q), \quad (7.15)$$

$$Q > \Lambda_s B \implies w_\xi^-(Q) \geq \underline{\mu}_\rho(Q) - \mathcal{L}_\rho(Q) \geq \delta_\rho(Q). \quad (7.16)$$

Proof. Suppose, for contradiction, that Q is the least unavailable prime-power step.

If $B < Q \leq \Lambda_s B$, all earlier composite losses total at most $\mathcal{L}_\rho(Q)$, while discarding every positive prime gain gives

$$w_\xi^-(Q) \geq L_\rho(B) - \mathcal{L}_\rho(Q) \geq \Delta_\rho(\Lambda_s B) \geq \delta_\rho(Q),$$

contradicting unavailability.

Now suppose $Q > \Lambda_s B$. For every $q \in \text{Prev}_\rho(Q)$ one has $B < q < Q$. By minimality of Q , the q -step has already occurred. Discarding every positive prime gain except the gain at q , and charging all composite losses up to Q , yields

$$w_\xi^-(Q) \geq \mu_\rho(q) - \mathcal{L}_\rho(Q).$$

Taking the minimum over $q \in \text{Prev}_\rho(Q)$ and using (7.12) gives (7.16), again a contradiction. Hence no unavailable step exists. $\square$

Successive intervals overlap. Their upper endpoints are unbounded because the positive prime gains are unbounded along the primes. Therefore the union of the intervals after the base contains a final ray in $\mathbb{N}$.

7.3. The affine centre and cutoff incidence. For $X \geq B$ let

$$\pi_{B,X} : A/E_B \longrightarrow A/E_X, \quad \gamma_X^\xi = \pi_{B,X}(\gamma_B^\xi).$$

For $B \leq X \leq Y$, write $\pi_{X,Y} : A/E_X \rightarrow A/E_Y$ for the further quotient. Functoriality of quotient maps is the commuting triangle

$$\begin{array}{ccc} A/E_B & \xrightarrow{\pi_{B,X}} & A/E_X \\ & \searrow \pi_{B,Y} & \downarrow \pi_{X,Y} \\ & & A/E_Y. \end{array} \quad (7.17)$$

Thus $\gamma_Y^\xi = \pi_{X,Y}(\gamma_X^\xi)$. Equation (2.27) gives at least one X with $\gamma_X^\xi = 0$, and (7.17) then gives the same conclusion for every larger Y . Hence

$$\mathcal{V}_\xi = \{X \geq B : \gamma_X^\xi = 0\}$$

is a nonempty final segment.

Let I_X^ξ be the scalar interval after all prime-power steps of rank at most X . Define the cutoff-count incidence object

$$\mathfrak{X}_\xi = \{(k, X) \in \mathbb{N}_{>0} \times \mathbb{N} : X \geq B, X \in \mathcal{V}_\xi, k \in I_X^\xi\}, \quad (7.18)$$

with projection $p_\xi(k, X) = k$, and define

$$\mathfrak{R}_\xi = \{k_0 \in \mathbb{N}_{>0} : [k_0, \infty) \cap \mathbb{N}_{>0} \subseteq \text{im}(p_\xi)\}. \quad (7.19)$$

The preceding continuation and centre-vanishing statements imply $\mathfrak{R}_\xi \neq \emptyset$.

Define

$$\mathfrak{G}_{\text{sc}}(\eta, \sigma, \rho) = \{(B, \xi) : \text{Cont}_\rho(B), \xi \in \mathfrak{B}_B(\eta, \sigma, L_\rho)\}. \quad (7.20)$$

Theorem 7.2 (Eventual count coverage). *For every admissible (η, σ, ρ) , the B -projection*

$$\mathfrak{G}_{\text{sc}}(\eta, \sigma, \rho) \longrightarrow \mathbb{N}$$

has cofinite image. For every (B, ξ) in this object, the cutoff-count projection p_ξ has cofinite image, equivalently $\mathfrak{R}_\xi \neq \emptyset$.

Proof. Both $\text{Cont}_\rho(B)$ and the nonemptiness of $\mathfrak{B}_B(\eta, \sigma, L_\rho)$ hold for all sufficiently large B . This proves the first assertion. Fix such a prepared start. Lemma 7.1 gives overlapping intervals with unbounded upper endpoints, and $\mathcal{V}_\xi$ is a final segment. Hence

$$\text{im}(p_\xi) = \bigcup_{X \in \mathcal{V}_\xi} (I_X^\xi \cap \mathbb{N}_{>0})$$

contains a final ray. $\square$

8. FINITE PHYSICAL REALIZATION

This section is where simultaneous geometric compatibility is imposed. After a finite cutoff X has been fixed, Haxell's theorem produces a common independent family over the finitely many active prime-power rows. On that family the path-label map factors through the joint-compatibility domain, so the compatible covering theorem pastes the local transfers to a covering physical correspondence.

Let

$$0 < \eta < \eta_0, \quad 0 < \sigma < 1, \quad 0 < \rho < \kappa\sigma\rho_P$$

be arbitrary parameters as in Sections 5–7. Define the reciprocal-mass and packing cutoff objects

$$\mathcal{B}_{\text{mass}}(\eta) = \left\{ B : \sum_{Q>B} \omega_Q < \eta \right\},$$

$$\mathcal{B}_{\text{pack}}(\sigma) = \left\{ B : \kappa\sigma \frac{Q}{\log Q} \geq L_P \text{ for every prime power } Q > B \right\}.$$

By (4.16) and $Q/\log Q \rightarrow \infty$, both contain final segments of $\mathbb{N}$. The physical cutoff object is their intersection with the continuation object:

$$\mathcal{B}_{\text{phys}}(\eta, \sigma, \rho) = \mathcal{B}_{\text{mass}}(\eta) \cap \mathcal{B}_{\text{pack}}(\sigma) \cap \mathcal{B}_{\text{cont}}(\rho). \quad (8.1)$$

It too contains a final segment. Thus there is a common onset $B_*(\eta, \sigma, \rho)$ beyond which the continuation, packing and reciprocal-mass requirements all hold.

Let $B \in \mathcal{B}_{\text{phys}}(\eta, \sigma, \rho)$ and let

$$\xi \in \mathfrak{W}_B(\eta, \sigma, L_\rho)$$

be arbitrary. The argument below is fibrewise in ξ . Write $R_B^\xi, \gamma_B^\xi, I_B^\xi$ for its initial data and $\mathcal{A}_Q(\xi)$ for its surviving later families.

8.1. The Haxell parameter set. For $X \geq B$, only finitely many prime-power steps lie between B and X . For every such Q , Proposition 5.2 gives

$$|\mathcal{A}_Q(\xi)| \geq \sigma |\mathcal{A}_Q| \geq \kappa\sigma \frac{Q}{\log Q}. \quad (8.2)$$

The classes $\mathcal{A}_Q(\xi)$ are pairwise disjoint by Lemma 4.2. Let $G_{X,\xi}$ be the incompatibility graph on their finite union, partitioned by these classes. By the definition of Δ_* in Section 4, the graph has maximum degree at most Δ_* . Define

$$\mathfrak{P}_{X,\xi} = \text{IQ}_{L_P}(G_{X,\xi}; (\mathcal{A}_Q(\xi))_{B < Q \leq X}). \quad (8.3)$$

By Proposition 3.3,

$$\mathfrak{P}_{X,\xi} \neq \emptyset. \quad (8.4)$$

Every $P \in \mathfrak{P}_{X,\xi}$ satisfies

$$|P \cap \mathcal{A}_Q(\xi)| \geq \rho_P |\mathcal{A}_Q(\xi)| > \rho \frac{Q}{\log Q}. \quad (8.5)$$

Hence

$$P(Q) := P \cap \mathcal{A}_Q(\xi) \in \mathcal{D}_\rho(Q)$$

for every prime-power step $B < Q \leq X$.

8.2. Local affine translation at one prime-power step. At a prime-power step Q , put

$$H = F_{<Q}, \quad K = F_Q.$$

The general exact sequence (2.16) specializes to

$$0 \rightarrow S_Q \rightarrow A/F_{<Q} \rightarrow A/F_Q \rightarrow 0, \quad (8.6)$$

and

$$\text{Fib}_{\pi_{F_{<Q}, F_Q}}(0) = S_Q.$$

For $P \in \mathfrak{P}_{X, \xi}$ define the finite local witness object

$$\mathcal{O}_P(Q) = \{O \subseteq P(Q) : |O| \leq p\} \quad (8.7)$$

with label map

$$\lambda_{Q, P} : \mathcal{O}_P(Q) \rightarrow C, \quad O \mapsto \bigcup_{I \in O} I.$$

Because P is independent, every displayed union belongs to C . By (4.10), its residue lies in F_Q , and its image in $S_Q \times \mathbb{N}$ is exactly the point of

$$\Gamma_{\phi_Q|_{P(Q)}}$$

represented by the same subset O . Thus Theorem 6.2 states precisely that the map (2.22), with $\beta = 0$, is a regular epimorphism onto the scalar target rectangle at this step. Corollary 2.7 therefore turns the abstract quotient support into a covering labelled correspondence.

Write

$$B < Q_1 < \cdots < Q_m \leq X$$

for the possibly empty list of processed prime-power steps.

Lemma 8.1 (Endpoint ladder). *If $m > 0$, then*

$$E_B = F_{<Q_1}, \quad E_{Q_j} = F_{Q_j} = F_{<Q_{j+1}} \quad (1 \leq j < m), \quad E_X = F_{Q_m}.$$

Consequently the quotient used by the Q_j -local affine translation is exactly the consecutive endpoint quotient from the preceding target to the next one.

Proof. By definition Q_1 is the first prime-power rank strictly larger than B , so the prime-power stages of rank at most B are exactly those of rank $< Q_1$, giving $E_B = F_{<Q_1}$. Likewise no prime-power rank lies strictly between consecutive Q_j, Q_{j+1} , hence $F_{Q_j} = F_{<Q_{j+1}}$. Finally no prime-power rank lies in $(Q_m, X]$, so $E_X = F_{Q_m}$. The identities $E_{Q_j} = F_{Q_j}$ are (2.28). $\square$

The successive affine translations therefore paste along the endpoint ladder

$$E_B = F_{<Q_1} \rightarrow E_{Q_1} = F_{Q_1} \rightarrow \cdots \rightarrow E_{Q_m} = F_{Q_m} = E_X, \quad (8.8)$$

with $F_{Q_j} = F_{<Q_{j+1}}$ at every intermediate junction.

Put

$$T_0^\xi = \text{Fib}_{E_B}(\gamma_B^\xi) \times I_B^\xi,$$

and for $1 \leq j \leq m$ let

$$\gamma_j^\xi = \pi_{B, Q_j}(\gamma_B^\xi), \quad T_j^\xi = \text{Fib}_{E_{Q_j}}(\gamma_j^\xi) \times I_{Q_j}^\xi.$$

For each $P \in \mathfrak{P}_{X,\xi}$ and $1 \leq j \leq m$, Corollary 2.7 gives a covering labelled correspondence

$$\mathcal{T}_{j,P} : T_{j-1}^\xi \rightsquigarrow T_j^\xi. \quad (8.9)$$

The filtration and scalar interval change only at prime-power steps, so $T_m^\xi = T_X^\xi$ when $m > 0$, while $T_0^\xi = T_X^\xi$ when $m = 0$.

8.3. Joint compatibility and strict pasting.

Lemma 8.2 (Joint compatibility). *For every $P \in \mathfrak{P}_{X,\xi}$, the product label map formed from an edge label of R_B^ξ and arbitrary local witnesses*

$$O_j \in \mathcal{O}_P(Q_j) \quad (1 \leq j \leq m)$$

lands in D_{m+1} .

Proof. All interval components occurring in the local labels belong to the single independent set P , hence are pairwise disjoint and nonadjacent. Each belongs to some $\mathcal{A}_{Q_j}(\xi)$ and is therefore compatible with every component in the prepared-start pool $\mathcal{P}_B^\xi$, which contains every component of every label of R_B^ξ . Thus every product label is jointly compatible and its literal union lies in C . $\square$

Put $n = m + 1$. For $P \in \mathfrak{P}_{X,\xi}$ let $P_n(P)$ be the path pullback of the edge objects in

$$R_B^\xi, \mathcal{T}_{1,P}, \dots, \mathcal{T}_{m,P}.$$

Lemma 8.2 gives the factorization

$$P_n(P) \xrightarrow{\tilde{\ell}_P} D_n \longrightarrow C^n, \quad \ell_P = \iota \circ \tilde{\ell}_P. \quad (8.10)$$

Hence the compatible path pullback is canonically isomorphic to the full path pullback, and (2.14) gives

$$U(R_{X,\xi,P}) = U(\mathcal{T}_{m,P}) \circ \dots \circ U(\mathcal{T}_{1,P}) \circ U(R_B^\xi), \quad (8.11)$$

where

$$R_{X,\xi,P} := \mathcal{T}_{m,P} \odot \dots \odot \mathcal{T}_{1,P} \odot R_B^\xi : \{0_M\} \rightsquigarrow T_X^\xi. \quad (8.12)$$

Every factor is covering, and (8.10) is precisely the remaining hypothesis of Theorem 2.4. That theorem makes $R_{X,\xi,P}$ covering by pullback stability. Equation (8.11) records the corresponding equality after passage to regular images.

Define the physical composite over the Haxell parameter set by the finite coproduct

$$R_{X,\xi}^{\text{univ}} = \coprod_{P \in \mathfrak{P}_{X,\xi}} R_{X,\xi,P}. \quad (8.13)$$

Because (8.4) is nonempty and every summand is covering, the target map

$$t_{X,\xi} : E_{R_{X,\xi}^{\text{univ}}} \rightarrow T_X^\xi \quad (8.14)$$

is a regular epimorphism.

8.4. Reciprocal-mass control. Every label of R_B^ξ has reciprocal value below $2 - \eta$. Every local label at step $Q = p^e$ is a compatible union of at most p members of $\mathcal{A}_Q$, so Proposition 4.3 bounds its reciprocal mass by ω_Q . Since $B \in \mathcal{B}_{\text{mass}}(\eta)$, (2.15) gives, uniformly in $P \in \mathfrak{P}_{X,\xi}$,

$$W(S) < 2 - \eta + \sum_{Q \geq B} \omega_Q < 2 \quad (8.15)$$

for every edge label S of $R_{X,\xi,P}$, and hence for every edge label of the coproduct (8.13).

Proposition 8.3 (Finite realization through a cutoff). *Let $B \in \mathcal{B}_{\text{phys}}(\eta, \sigma, \rho)$, let*

$$\xi \in \mathfrak{B}_B(\eta, \sigma, L_\rho), \quad X \geq B.$$

Then the Haxell parameter set $\mathfrak{P}_{X,\xi}$ is nonempty and the coproduct correspondence

$$R_{X,\xi}^{\text{univ}} : \{0_M\} \rightsquigarrow T_X^\xi, \quad T_X^\xi = \text{Fib}_{E_X}(\gamma_X^\xi) \times I_X^\xi,$$

is covering. Every edge label S satisfies $W(S) < 2$.

9. FINAL EXACTIFICATION

The final step is a base change followed by a regular-image factorization. It uses the arithmetic only through the covering correspondences, their mass bound and the cofinite count image already established.

Fix

$$\eta_* = \frac{\eta_0}{2}, \quad \sigma_* = \frac{1}{2}, \quad \rho_* = \frac{\kappa \rho_P}{4}.$$

These parameters satisfy all preceding hypotheses. Theorem 7.2 and the final-segment property of (8.1) supply a base B and a prepared start

$$\xi \in \mathfrak{B}_B(\eta_*, \sigma_*, L_{\rho_*}), \quad B \in \mathcal{B}_{\text{phys}}(\eta_*, \sigma_*, \rho_*).$$

Fix this pair and a threshold $k_0 \in \mathfrak{R}_\xi$. Thus, for every $k \geq k_0$, some finite cutoff $X \geq B$ satisfies

$$\gamma_X^\xi = 0, \quad k \in I_X^\xi.$$

Put $J = [k_0, \infty) \cap \mathbb{N}$ and form the cutoff incidence set

$$\mathfrak{X} = \{(k, X) : k \in J, X \geq B, \gamma_X^\xi = 0, k \in I_X^\xi\}. \quad (9.1)$$

The projection

$$p : \mathfrak{X} \twoheadrightarrow J, \quad p(k, X) = k, \quad (9.2)$$

is a regular epimorphism by the defining property of k_0 .

For every integer $X \geq B$, Proposition 8.3 gives a covering correspondence

$$R_X := R_{X,\xi}^{\text{univ}} : \{0_M\} \rightsquigarrow T_X^\xi.$$

The next coproduct is only a set-theoretic packaging of the separate finite-cutoff correspondences; no composite across different cutoffs is being formed. Let $\mathcal{E} = \coprod_{X \geq B} E_{R_X}$ and $\mathcal{T} = \coprod_{X \geq B} T_X^\xi$. The target maps assemble to a regular

epimorphism $t : \mathcal{E} \twoheadrightarrow \mathcal{T}$. For $(k, X) \in \mathfrak{X}$, the point $(0, k)$ belongs to T_X^ξ . Pull back along these points:

$$\begin{array}{ccc} \mathfrak{E} & \longrightarrow & \mathcal{E} \\ e \downarrow & & \downarrow t \\ \mathfrak{X} & \xrightarrow{(k, X) \mapsto (X, (0, k))} & \mathcal{T}. \end{array} \quad (9.3)$$

Pullback stability makes e a regular epimorphism, so $pe : \mathfrak{E} \twoheadrightarrow J$ is a regular epimorphism as well. This construction requires no compatible choice of witnesses as X varies: each witness lies over one finite cutoff.

The label maps assemble to $\ell : \mathfrak{E} \rightarrow C$. If $u \in \mathfrak{E}$ lies over (k, X) , conservation and the mass bound give

$$\overline{W}(\ell(u)) = 0, \quad \nu(\ell(u)) = k > 0, \quad W(\ell(u)) < 2.$$

Let $q_{\mathbb{Z}} : \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z}$ be the quotient map. The commuting triangle

$$\begin{array}{ccc} C & \xrightarrow{W} & \mathbb{Q} \\ & \searrow \overline{W} & \downarrow q_{\mathbb{Z}} \\ & & \mathbb{Q}/\mathbb{Z} \end{array} \quad (9.4)$$

puts $W(\ell(u))$ in $\ker q_{\mathbb{Z}} = \mathbb{Z}$. Since the label is nonempty, its reciprocal value is positive. Thus $W(\ell(u)) \in (0, 2) \cap \mathbb{Z} = \{1\}$, and the map $u \mapsto (pe(u), \ell(u))$ factors through the solution set:

$$\begin{array}{ccc} \mathfrak{E} & \xrightarrow{u \mapsto (pe(u), \ell(u))} & \mathcal{S} \\ pe \downarrow & & \downarrow \pi_{\text{sol}} \\ J & \longrightarrow & \mathbb{N}_{>0}. \end{array} \quad (9.5)$$

Taking regular images gives $J \subseteq \text{im}(\pi_{\text{sol}})$. Since J is cofinite, this proves Theorem 1.1.

Proof of Corollary 1.2. Keep the same base B and prepared start ξ for every $k \geq k_0$. The seed and neutral switches use only two-element components. Every three-element component of a prefix label therefore belongs to the finite pool of bridge alternatives through B . All later local labels are unions of two-element intervals, and compatible pasting cannot merge components.

Let $\mathcal{T}_B$ be the finite set of all three-element connected components occurring in a prefix label $\lambda_B(b)$ with $b \in \widetilde{\mathcal{B}}_B$. Put $C = \max\{1, |\mathcal{T}_B|\}$, and take $M \geq 1$ containing all their support points. Every solution obtained above has at most C three-element components, all contained in $[1, M]$. If their number is t , then $|S| = 2\nu(S) + t = 2k + t$, which proves the corollary. $\square$

APPENDIX A. COMPARABLE PRIME SUPPLY

This appendix proves the ordinary analytic input used in Sections 4 and 7: one fixed-ratio prime band containing $\asymp X/\log X$ primes. We take the multiplier to be the least admissible integer, so every later cutoff remains on the natural-number scale.

Put

$$\theta(x) = \sum_{p \leq x} \log p, \quad \psi(x) = \sum_{p^m \leq x} \log p = \log \text{lcm}(1, \dots, \lfloor x \rfloor).$$

The central binomial coefficient satisfies

$$\binom{2n}{n} \mid \text{lcm}(1, \dots, 2n+1).$$

Indeed, for each prime p , Legendre's formula gives

$$v_p \binom{2n}{n} = \sum_{j \geq 1} \left(\left\lfloor \frac{2n}{p^j} \right\rfloor - 2 \left\lfloor \frac{n}{p^j} \right\rfloor \right),$$

and every summand is 0 or 1; if the sum is e , then some nonzero summand has index at least e , so $p^e \leq 2n$.

Since the central binomial coefficient is maximal in its row,

$$4^n \leq (2n+1) \binom{2n}{n}.$$

The divisibility above therefore gives

$$\psi(2n+1) \geq \log \binom{2n}{n} \geq n \log 4 - \log(2n+1).$$

Taking $n = \lfloor (x-1)/2 \rfloor$ and using monotonicity of ψ shows that, for some $c_\psi > 0$,

$$\psi(x) \geq c_\psi x$$

for all sufficiently large x .

On the other hand, every prime $n < p \leq 2n$ divides $\binom{2n}{n}$, so

$$\theta(2n) - \theta(n) \leq \log \binom{2n}{n} \leq 2n \log 2.$$

Dyadic telescoping gives $\theta(x) \leq C_\theta x$ for a fixed C_θ . Since

$$\psi(x) = \sum_{m \geq 1} \theta(x^{1/m}),$$

there are only $O(\log x)$ nonzero terms with $m \geq 2$, and the upper bound for θ gives

$$0 \leq \psi(x) - \theta(x) = \sum_{m \geq 2} \theta(x^{1/m}) = O(\sqrt{x} \log x) = o(x).$$

Subtracting this from the lower bound for ψ yields, after increasing the threshold if necessary,

$$c_\theta x \leq \theta(x) \leq C_\theta x \tag{A.1}$$

for some $c_\theta > 0$.

The set

$$\{m \in \mathbb{N} : m \geq 2, C_\theta < c_\theta m\}$$

is nonempty. Let Λ be its least element. Then

$$\theta(\Lambda X) - \theta(X) \geq (c_\theta \Lambda - C_\theta)X,$$

while every prime in $(X, \Lambda X]$ contributes between $\log X$ and $\log(\Lambda X)$. Thus:

Proposition A.1 (Comparable prime band). *There are an integer $\Lambda \geq 2$ and constants $c, C > 0$ such that, for every sufficiently large integer X ,*

$$c \frac{X}{\log X} \leq \#\{p : X < p \leq \Lambda X\} \leq C \frac{X}{\log X}.$$

Only the existence of fixed constants Λ, c, C is used in the main argument; their numerical values play no role.

APPENDIX B. FIXED-CARDINALITY SUMS IN A PRIME CYCLIC GROUP

The theorem below is due to Dias da Silva and Hamidoune [6]. Alon, Nathanson and Ruzsa later recovered it as a consequence of their polynomial method [2]. We include a self-contained polynomial proof tailored to the form used in Section 6. For a finite set A , write $\binom{A}{h}$ for the set of its h -element subsets.

Theorem B.1 (Dias da Silva–Hamidoune). *Let S be a cyclic group of prime order p , let $A \subseteq S$ with $|A| = r$, and let $0 \leq h \leq r$. Define*

$$\Sigma_h : \binom{A}{h} \longrightarrow S, \quad O \longmapsto \sum_{a \in O} a.$$

Then

$$|\operatorname{im} \Sigma_h| \geq \min\{p, h(r-h) + 1\}.$$

In particular, if $h(r-h) + 1 \geq p$, then Σ_h is surjective.

Proof. The cases $h = 0, r$ are immediate, so assume $0 < h < r$. The assertion is invariant under group isomorphism, so it suffices to prove it for $S = \mathbb{F}_p$. Thus $A \subseteq \mathbb{F}_p$ and $r \leq p$.

Put

$$M = \min\{p-1, h(r-h)\}.$$

Suppose for contradiction that $|\operatorname{im} \Sigma_h| \leq M$. Enlarge the image to a set $C \subseteq \mathbb{F}_p$ of cardinality exactly M and consider

$$P(x_1, \dots, x_h) = \prod_{c \in C} \left(\sum_{i=1}^h x_i - c \right) \prod_{1 \leq i < j \leq h} (x_j - x_i). \quad (\text{B.1})$$

The polynomial vanishes on A^h : repeated coordinates kill the Vandermonde factor, while distinct coordinates have sum in $\operatorname{im} \Sigma_h \subseteq C$.

We use two coefficient facts. First, if $F \in K[x_1, \dots, x_h]$ has total degree at most $\sum_i d_i$, if

$$[x_1^{d_1} \cdots x_h^{d_h}] F \neq 0,$$

and if finite sets A_i satisfy $|A_i| > d_i$, then F is nonzero somewhere on $\prod_i A_i$. This is the coefficient form of the combinatorial Nullstellensatz [1], and follows directly from the one-variable Lagrange coefficient-extraction formula.

Second, for distinct nonnegative integers $k_1, \dots, k_h$ and

$$m = \sum_i k_i - \frac{h(h-1)}{2},$$

one has

$$[x_1^{k_1} \cdots x_h^{k_h}] (x_1 + \cdots + x_h)^m \prod_{i < j} (x_j - x_i) = m! \frac{\prod_{i < j} (k_j - k_i)}{\prod_i k_i!}. \quad (\text{B.2})$$

Indeed, expanding the Vandermonde determinant reduces the alternating sum to a determinant of falling factorials, hence to the ordinary Vandermonde determinant.

For every $0 \leq m \leq h(r-h)$ the following construction gives distinct

$$0 \leq k_1 < \cdots < k_h \leq r-1$$

with

$$\sum_i k_i = m + \frac{h(h-1)}{2}. \quad (\text{B.3})$$

Write $R = r-h$ and $m = qh + s$ with $0 \leq s < h$. Define

$$\lambda_i = \begin{cases} q, & 1 \leq i \leq h-s, \\ q+1, & h-s < i \leq h, \end{cases} \quad k_i = i-1 + \lambda_i.$$

If $q = R$, then $m \leq hR$ forces $s = 0$; hence always $0 \leq \lambda_i \leq R$. The sequence (λ_i) is nondecreasing, so the k_i are strictly increasing and lie in $[0, r-1]$.

It remains to connect (B.2) with the polynomial (B.1). Put

$$H_C(T) = \prod_{c \in C} (T - c), \quad V(x) = \prod_{i < j} (x_j - x_i).$$

Since $|C| = M$, the polynomial H_C is monic of degree M :

$$H_C(T) = T^M + R_C(T), \quad \deg R_C < M. \quad (\text{B.4})$$

The Vandermonde is homogeneous of degree $h(h-1)/2$. Hence every monomial of

$$R_C(x_1 + \cdots + x_h)V(x)$$

has total degree strictly below $M + h(h-1)/2$. Therefore, for every multi-index k of that total degree,

$$[x^k]P = [x^k](x_1 + \cdots + x_h)^M V(x). \quad (\text{B.5})$$

Thus the lower homogeneous terms of H_C cannot contribute to the coefficient used below.

Apply (B.3) with $m = M$. By (B.5) and (B.2), the corresponding coefficient of P is

$$M! \frac{\prod_{i < j} (k_j - k_i)}{\prod_i k_i!}. \quad (\text{B.6})$$

It is nonzero in $\mathbb{F}_p$: $M \leq p-1$ and $0 \leq k_i \leq r-1 \leq p-1$, so the factorials are nonzero, and the distinct $k_i \in \{0, \dots, p-1\}$ have pairwise differences nonzero modulo p . The coefficient criterion contradicts the vanishing of P on A^h .

Hence $|\text{im } \Sigma_h| > M$, which is equivalent to the required bound. $\square$

APPENDIX C. A FINITE INITIALIZATION

This appendix supplies one explicit finite seed for the uniform construction of Section 5; no property of the particular numerical values is used later.

Put

$$P(m) = \frac{1}{m} + \frac{1}{m+1}.$$

Lemma C.1 (Finite initialization). *There is a finite family of configurations in C with the following properties.*

- (1) *Their residues exhaust the zero-centred H_6 -fibre.*
- (2) *Their component counts have a finite spread independent of every later prime-power step.*
- (3) *They are compatible with the finitely many exceptional bridge stages preceding the uniform separation regime.*
- (4) *The reciprocal value of every initialized branch, together with the whole subsequent bridge sequence, is bounded by a constant strictly below 2.*

Proof. We first give an exact finite witness for the first three assertions, and then prove the uniform separation and reciprocal-mass estimates needed for the fourth.

One explicit witness.

The empty configuration gives residue 0. The following table gives five further configurations; the third column lists the starts of their binary components $[m, m + 1]$.

residue	component count	component starts
$\frac{1}{6}$	9	35, 56, 90, 119, 152, 170, 455, 494, 714
$\frac{1}{3}$	15	17, 35, 44, 84, 104, 119, 132, 170 272, 285, 455, 494, 560, 714, 935
$\frac{1}{2}$	7	9, 17, 34, 44, 77, 84, 90
$\frac{2}{3}$	15	7, 16, 34, 44, 51, 65, 77, 119 132, 153, 170, 272, 285, 455, 494
$\frac{5}{6}$	1	2

(C.1)

Equivalently,

$$\frac{j}{6} = \sum_{m \in M_j} P(m) \quad (j = 1, 2, 3, 4, 5),$$

where M_j is the corresponding row of the table. The following common-denominator certificate records the exact arithmetic. Writing

$$L_j = \text{lcm}_{m \in M_j} m(m + 1), \quad N_j = \sum_{m \in M_j} (2m + 1) \frac{L_j}{m(m + 1)},$$

one obtains

j	L_j	N_j	N_j/L_j
1	116396280	19399380	1/6
2	232792560	77597520	1/3
3	3063060	1531530	1/2
4	232792560	155195040	2/3
5	6	5	5/6

(C.2)

Thus the five rational identities are exact. The minimum difference between consecutive starts in the five rows of (C.1) is respectively 18, 9, 6, 7 (and vacuous in the last singleton row), so every row belongs to C . Together with the empty configuration the six residues exhaust H_6 .

The two quotient jumps before the uniform bridge regime are

$$6 \longrightarrow 12, \quad 12 \longrightarrow 60.$$

For $p = 2$ use coefficient set $\{1\}$, giving bridge base 12. For $p = 5$ use $\{1, 2, 3\}$, giving bases 60, 30, 20. Their subset sums cover $\mathbf{Z}/2\mathbf{Z}$ and $\mathbf{Z}/5\mathbf{Z}$; for $p = 5$, the residues 0, 1, 2, 3, 4 are represented by

$$\emptyset, \{1\}, \{2\}, \{3\}, \{1, 3\}.$$

The bridge identity is exactly (5.1): at lower modulus D and quotient prime p , the alternatives at base $a = pD/r$ differ in reciprocal value by $r/(pD)$ and have the same component count.

Only a finite boundary remains before the separation estimate becomes automatic. One sufficient list of bridge bases is

$$12, 20, 30, 60, 140, 210, 420, 840, 1260, 2520. \quad (\text{C.3})$$

For completeness, the finite separation check can also be recorded numerically. Against the worst bridge alternative $[a, a+2]$ with a in (C.3), the minimum number of unused integers between that interval and a seed component is respectively

$$2, 1, 1, 1, 8$$

for the five nonzero residue rows of (C.1); between two bridge alternatives from the displayed list the minimum is 5. Hence all simultaneously occurring seed and bridge components are disjoint and nonadjacent. The list is part of this particular finite witness, not an intrinsic sequence.

Why the numerical initialization disappears.

Suppose a previous prime-power jump has upper least-common-multiple modulus D . Every bridge base in that stage is at most D , so every component support ends by $D + 2$ and its exclusion envelope ends by $D + 3$. At the next nontrivial jump the lower modulus is the same D , the quotient prime is p , and the smallest new base is

$$a_k = \frac{pD}{k},$$

where k is least with $k(k+1)/2 \geq p-1$. Thus

$$a_k - D = D \left(\frac{p}{k} - 1 \right). \quad (\text{C.4})$$

In fact

$$\frac{p}{k} - 1 \geq \frac{1}{2}$$

for every prime p . The cases $p = 2, 3, 5$ are immediate. For $p \geq 7$, put $m = \lfloor 2p/3 \rfloor$. Since $m \geq 2p/3 - 1$,

$$\frac{m(m+1)}{2} \geq \frac{2p^2}{9} - \frac{p}{3} \geq p-1,$$

so minimality gives $k \leq m \leq 2p/3$.

The bases in (C.3) are exactly those needed through the jump at $n = 9$. After that jump the next lower modulus is

$$D = L_{10} = 2520.$$

Hence every subsequent stage satisfies

$$a_k - D \geq D/2 \geq 1260.$$

The new component supports therefore begin far beyond the preceding exclusion envelopes as well as the fixed finite seed, and the same estimate persists inductively as the lower moduli increase. Within each new row, minimality also gives

$$k(k-1) < 2(p-1), \quad k \leq p-1, \quad k^2 < 3p.$$

Consequently (5.2) gives $a_r - a_s > D/3 \geq 840$ for $r < s$ in every subsequent row. Both within-row and between-row separation therefore hold throughout the uniform regime.

The reciprocal-mass estimate is likewise uniform. The largest initial reciprocal value is $5/6$. For the four finite bridge bases 12, 20, 30, 60, the crude reciprocal bound $< 3/a$ gives

$$\frac{5}{6} + \frac{3}{12} + \frac{3}{20} + \frac{3}{30} + \frac{3}{60} = \frac{5}{6} + \frac{11}{20} < \frac{3}{2}. \quad (\text{C.5})$$

For every later bridge, with lower modulus D and k least such that $T_k = k(k+1)/2 \geq p-1$, minimality gives $T_k < 2p$. At base $a_r = pD/r$, every alternative has reciprocal mass below $3/a_r$, so an entire branch has mass

$$< \sum_{r=1}^k \frac{3r}{pD} = \frac{3T_k}{pD} < \frac{6}{D}. \quad (\text{C.6})$$

After the finite initialization the first lower modulus is at least 60, and successive nontrivial least-common-multiple jumps multiply the lower modulus by at least two. Hence

$$\sum_{\text{later bridge stages}} \text{reciprocal mass} < \sum_{j \geq 0} \frac{6}{60 \cdot 2^j} = \frac{1}{5} < \frac{1}{4}. \quad (\text{C.7})$$

Together with (C.5) this leaves a fixed positive margin below 2 before the arbitrarily small neutral contribution is added. This verifies all four assertions. $\square$

Thus the main argument uses only Lemma C.1. The displayed residue representatives, bridge bases and finite boundary list play no further role.

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